MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, HIGH SPEED, CMOS
LATCHES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, high speed, CMOS, latches, bistable logic microcircuits. Two product assurance classes and a choice of case outlines and lead finish are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device types shall be as follows:

Device type	Circuit
01	Qual, bistable, transparent latch
02	8-Bit addressable latch
03	Octal, D type latch, 3-state
04	Octal transparent latch, 3-state, inverting
05	Octal transparent latch, 3-state inverting
06	Octal transparent latch, 3-state
53	Octal D-type flip-flop, 3-state and TTL-compatible inputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16 lead, 1/4" x 3/4"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
2	C-2 (20-terminal, .350" x .350") square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage	-0.5 V dc to +7.0 V dc
DC input voltage	-0.5 V dc to V_{CC} +1.5 V dc
DC output voltage	-0.5 V dc to V_{CC} +1.5 V dc
Clamp diode current	±20 mA
DC current per pin	±25 mA (±35 mA for type 03-06, 53)
DC V_{CC} or GND current per pin	±50 mA (±70 mA for type 03-06, 53)
Storage temperature range	-65°C to +150°C
Maximum power dissipation	300mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases E, F, R, and S	(See MIL-M-38510, appendix C)
Case 2	50°C/W
Junction temperature (T_J)	+175°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions for device 01 - 06.

Input low (V_{IL}) maximum voltage	0.3 V at V_{CC} = 2 V 0.9 V at V_{CC} = 4.5 V 1.2 V at V_{CC} = 6 V
Input high (V_{IH}) minimum voltage	1.5 V at V_{CC} = 2 V 3.15 V at V_{CC} = 4.5 V 4.2 V at V_{CC} = 6 V
Supply voltage (V_{CC})	2 V dc to 6 V dc
Input voltage	0 V dc to V_{CC}
Operating temperature	-55°C to +125°C

Input rise and fall times (t_r, t_f) maximum:

V_{CC} = 2.0 V	1000 ns
V_{CC} = 4.5 V	500 ns
V_{CC} = 6.0 V	400 ns

Device 53

Input low (V_{IL}) maximum voltage	0.8 V at V_{CC} = 4.5 to 5.5 V dc
Input high (V_{IH}) minimum voltage	2.0 V at V_{CC} = 4.5 to 5.5 V dc
Supply voltage (V_{CC})	4.5 V to 5.5 V dc
Output voltage	0 V dc to V_{CC}
Operating temperature	-55°C to +125°C
Input rise and fall times (t_r, t_f) at V_{CC} = 4.5 V:	500 ns max
Width of clock pulse (t_p clock) (V_{CC} = 4.5 V minimum):	
Devices 01, 02, 05, 06,	24 ns minimum
Devices 03, 04, 53	30 ns minimum
Device 02 reset	24 ns minimum
Data setup time before clock (t_{setup}): (V_{CC} = 4.5 V minimum):	
Devices 01-04, 53	30 ns minimum
Devices 05, 06	23 ns minimum
t_{HOLD}: (V_{CC} = 4.5 V minimum):	
Device types 01, 02	8 ns minimum
Device types 03-06, 53	15 ns minimum

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitaiton.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.6.1. Total dose radiation hardness identifier. The total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 38 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

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a. Delete the sequence specified as initial (preburn-in) electrical parameters through interim (postburn-in) electrical parameters of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 3, or equivalent. Ambient temperature (T_A) shall be +125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 3, or equivalent. Ambient temperature shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_i and C_o), measurements shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and GND at a frequency of 1 MHz.

d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified		Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH1} 3/	V_{IH} = 1.5 V V_{IL} = 0.3 V I_{OH} = -20 µA		01-06	2.0 V	1.95		V
	V_{OH2} 3/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OH} = -20 µA		01-06	4.5 V	4.45		V
	V_{OH3}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OH} = -20 µA		01-06	6.0 V	5.95		V
	V_{OH4} 3/	V_{IH} = 3.15 V V_{IL} = 0.9 V	I_{OH} = -6.0 mA	03-06	4.5 V	3.7		V
			I_{OH} = -4.0 mA	01,02				
	V_{OH5}	V_{IH} = 4.2 V V_{IL} = 1.2 V	I_{OH} = -5.2 mA	01,02,	6.0 V	5.2		V
			I_{OH} = -7.8 mA	03-06				
	V_{OH6}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OH} = -20 µA		53	4.5 V	4.4		V
	V_{OH7}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OH} = -6.0 mA		53	5.5 V	3.7		V
Low-level output voltage	V_{OL1} 3/	V_{IL} = 0.3 V V_{IH} = 1.5 V I_{OH} = 20.0 µA		01-06	2.0 V		0.05	V
	V_{OL2} 3/	V_{IL} = 0.9 V V_{IH} = 3.15 V I_{OH} = 20.0 µA		01-06	4.5 V		0.05	V
	V_{OL3}	V_{IL} = 1.2 V V_{IH} = 4.2 V I_{OH} = 20.0 µA		01-06	6.0 V		0.05	V
	V_{OL4} 3/	V_{IL} = 0.9 V V_{IH} = 3.15 V I_{OH} = 4.0 mA	I_{OL} = 4.0 mA	01,02	4.5 V		0.4	V
			I_{OL} = 6.0 mA	03-06				

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified		Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
Low-level output voltage	V_{OL5}	V_{IL} = 1.2 V V_{IH} = 4.2 V	I_{OH} = 5.2 mA	01,02,	6.0 V		0.4	V
			I_{OH} = 7.8 mA	03-06				
	V_{OL6}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OL} = 20 µA		53	4.5 V		0.1	V
	V_{OL7}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OL} = 6.0 mA		53	5.5 V		0.4	V
Positive input clamp voltage	V_{IC+}	V_{CC} = GND I_{IN} = 1 mA T_C = +25°C		All	GND		1.5	V
Negative input clamp voltage	V_{IC-}	V_{CC} = Open I_{IN} = -1 mA T_C = +25°C		All	OPEN		-1.5	V
Input current low	I_{IL}	V_{IN} = GND		01-06	6.0 V		-0.1	µA
				53	5.5 V			
Input current high	I_{IH}	V_{IN} = V_{CC}		01-06	5.0 V		0.1	µA
				53	5.5 V			
Short circuit output current	I_{OS1} 3/	V_O = GND V_I = GND or V_{CC}		01-02	2.0 V	-2	-50	mA
				03-06			-60	
	I_{OS2} 3/			01-02	4.5 V	-15	-150	
				03-06			-165	
	I_{OS3} 3/			01-02	6.0 V	-25	-180	
				03-06			-210	
	I_{OS4}			01-02	4.0 V	-10	-120	
				03-06, 53			-135	
	I_{OS5} 3/			53	5.5 V	-25	-210	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
Supply current quiescent	I_{CC}	V_I = 6.0 V or GND	01	6.0 V		10	µA
			02-06			20	
	I_{CC}	V_I = 5.5 V or GND	53	5.5 V		20	
	I_{CCZ}	V_I = 6.0 V	03-06	6.0 V		10	
		V_I = 5.5 V	53	5.5 V			
Additional supply current quiescent per input pin (one unit load)	$I_{CC\Delta}$ 6/	V_{IL} = 0.8 V, I_O = 0 V_{IH} = 2.4 V Pin under test at 2.4 V, other pins at GND, 0.8 V, or V_{CC}	53	5.5 V		1.5	µA
Power dissipation capacitance	C_{PD} 2/ 3/	T_C = 25°C	01			46	pF
			02			33	
			03			51	
			53			53	
			04-06			50	
Three-state output leakage current low	I_{OZL} 7/	V_{OUT} = GND OC = V_{IH} 4/	03-06	6.0 V		-2.0	µA
			53	5.5 V			
Three-state output leakage current high	I_{OZH} 7/	V_{OUT} = V_{CC} OC = V_{IH} 4/	03-06	6.0 V		+2.0	µA
			53	5.5 V			
Input capacitance	C_{IN}	T_C = 25°C	All			10	pF
Three-state output capacitance	C_O		03-06, 53			20	
Propagation delay data to outputs 4/, 5/	t_{PHL1} t_{PLH1}	C_L = 50 pF ±10%	01	4.5 V	3	30	ns
			02		5	43	
			03		6	35	
			04		6	39	
			05,06		4	41	
			53		6	41	
Propagation delay enable or clock to outputs 4/, 5/	t_{PHL2} t_{PLH2}		01		4	34	
			02		7	47	
			03,53		6	41	
			04		5	41	
			05		4	41	
			06		4	41	
Propagation delay delay A to outputs 4/, 5/	t_{PHL3} t_{PLH3}		02		5	50	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
Propagation delay clear to outputs 4/, 5/	t_{PHL4}	C_L = 50 pF ±10%	02	4.5 V	5	36	
Output enable time 4/, 5/	t_{PZH}	C_L = 50 pF ±10% R_L = 1K ±10%	03		5	35	
			04		5	35	
	t_{PZL}		05,06		4	35	
			53		5	41	
Output disable time 4/, 5/	t_{PHZ}		03		5	35	
			04		5	35	
	t_{PLZ}		05,06		3	35	
			53		5	41	
Transition delay 4/, 5/	t_{THL}	C_L = 50 pF ±10%	01		3	20	
			02		3	20	
	t_{TLH}		03,04, 53		2	16	
			05		2	16	
			06		2	16	

1/ Complete terminal conditions shall be as specified in table III.
2/ Power dissipation capacitance (C_{PD}) per latch.
3/ Guaranteed but not tested.
4/ Tested at V_{CC} = 4.5 V at 125°C for sample testing and V_{CC} = 4.5 V and 25°C for screening. Guaranteed at other V_{CC} voltages and temperatures, see table IA and the exception in 4.4.1d.
5/ For propagation and transition delay times at V_{CC} = 2.0 V, increase limit by a factor of 5. For propagation and transition delay times at V_{CC} = 6.0 V, decrease limit by a factor of .85.
6/ I_{CC} + I_{CCA} = Total supply current.
7/ I_{OZL}: Set output to high state.
I_{OZH}: Set output to low state.

TABLE IA. Calculated dynamic values at -55/+25°C case temperature.

V_{CC}	T_C = °C	
	+125	-55°C/+25°C
2.0 V	5	5 x 0.75
4.5 V	1	0.75
6.0 V	0.85	0.85 x 0.75

NOTE: Normalized numbers (+125°C = 1). The 2.0 V and 6.0 V numbers are derived from their 4.5 V integer value (rounding off according 5/4).

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TABLE II. Burn-in and electrical test requirements.

Line no.	Applicable tests and MIL-STD-883 test method	Class S device 1/				Class B device 1/			
		Ref. par.	Table III subgroups 2/	Table IV delta limits 3/	Test circuit figure	Ref. par.	Table III subgroups 2/ 5/	Table IV delta limits 3/	Test circuit figure
1	Interim electrical parameters (method 5004)		1				1		
2	Static burn-in I (method 1015)	4.5.2	Req'd		3				
3	Same as line 1		1	Δ					
4	Static burn-in II (method 1015)	4.5.2	Req'd		3		Req'd	4/	3
5	Same as line 1	4.2d	1*	Δ		4.2d	1*	Δ	
6	Dynamic burn-in (method 1015)	4.2c 4.5.2	Req'd	1/	3		Not req'd		
7	Same line 1	4.2d	1	Δ					
8	Final electrical parameters (method 5005)		1*,2,3,7 8,9				1*,2,7, 9 4/		
9	Group A test requirements (method 5005)	4.4.1	1,2,3,4, 7,8,9, 10,11			4.4.1	1,2,3,4, 7,8,9, 10,11		
10	Group B end-point electrical parameters (method 5005)	4.4.1	†1,2,3,9, 10,11	Δ	3		†1		
11	Group C end-point electrical parameters (method 5005)					4.4.3	1,2	Δ	3
12	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3			4.4.4	1,2		

1/ Blank spaces indicate tests are not applicable.

2/ * indicates PDA applies to subgroup 1 (see 4.2.1).

3/ Δ indicates delta limit shall be required only on table III subgroup 1, where specified, and the delta values shall be computed with reference to the previous interim electrical parameters.

4/ The device manufacturer may at his option either complete subgroup 1 electrical parameter measurements, including delta measurements, within 96 hours after burn-in completion (removal of bias); or may complete subgroup 1 electrical measurements without delta measurements within 24 hours after burn-in completion (removal of bias).

5/ † indicates also applies to electrostatic discharge sensitivity tests.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady-state life test circuits shall be submitted to the qualifying activity for approval. When the alternate steady-state life test is used, the circuit on figure 3 (dyanmic) or equivalent, shall be used.

b. Electrostatic discharge sensitivity (ESDS) testing shall be performed in accordance with MIL-STD-883, method 3015. The option to categorize devices as ESD sensitive without performing the test is not allowed. Device types categorized as ESD sensitive shall be further tested using method 3015 modified as follows:

1. Test method 3015, table I pin combinations 4 and 5 shall be deleted.

2. The test sequence specified in paragraph 3.b shall be repeated an additional four times rather than the two specified.

3. The category A limits specified in figure 3015-3 ESD sensitivity category shall be 1000 V to 2000 V.

c. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 5 of group B inspections for class S, and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 herein and as shown on figure 3 (note 3), or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

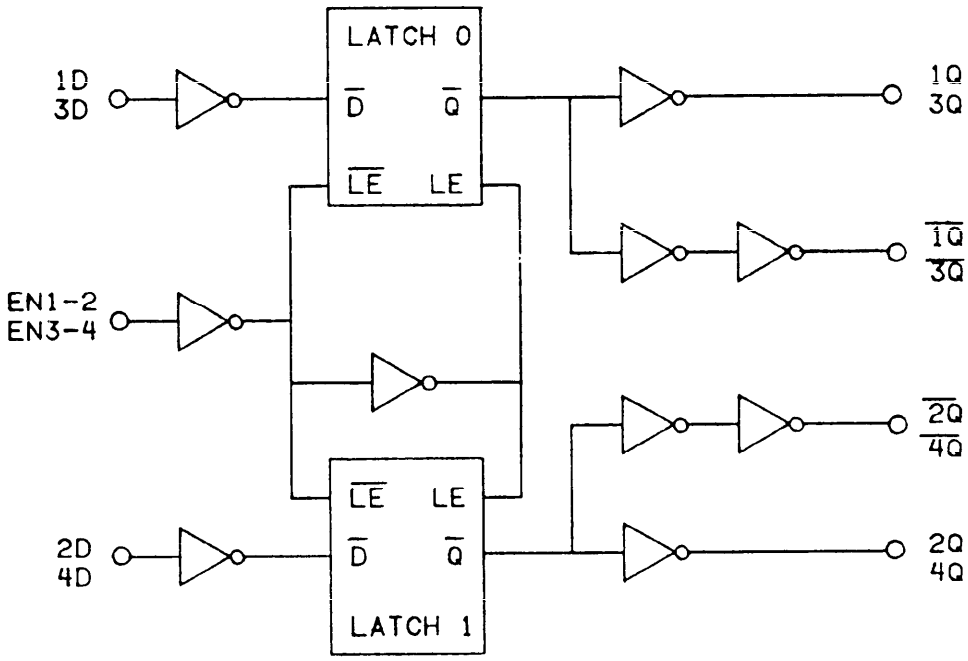
4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed, it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.4 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit GND terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

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Device type 01

Cases E and F

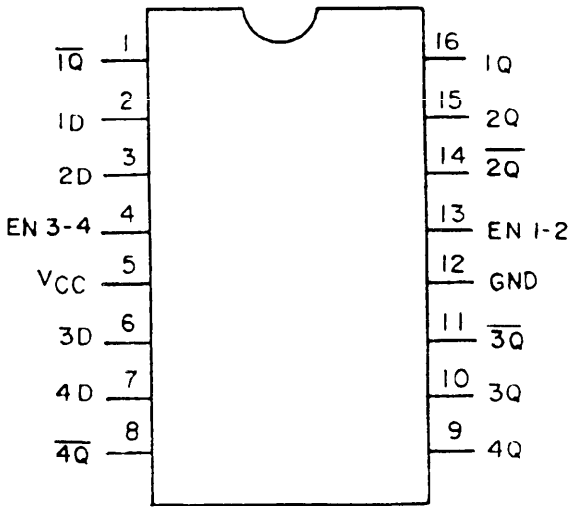


FIGURE 1. Logic diagrams and terminals connections.

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Device type 01

Case 2

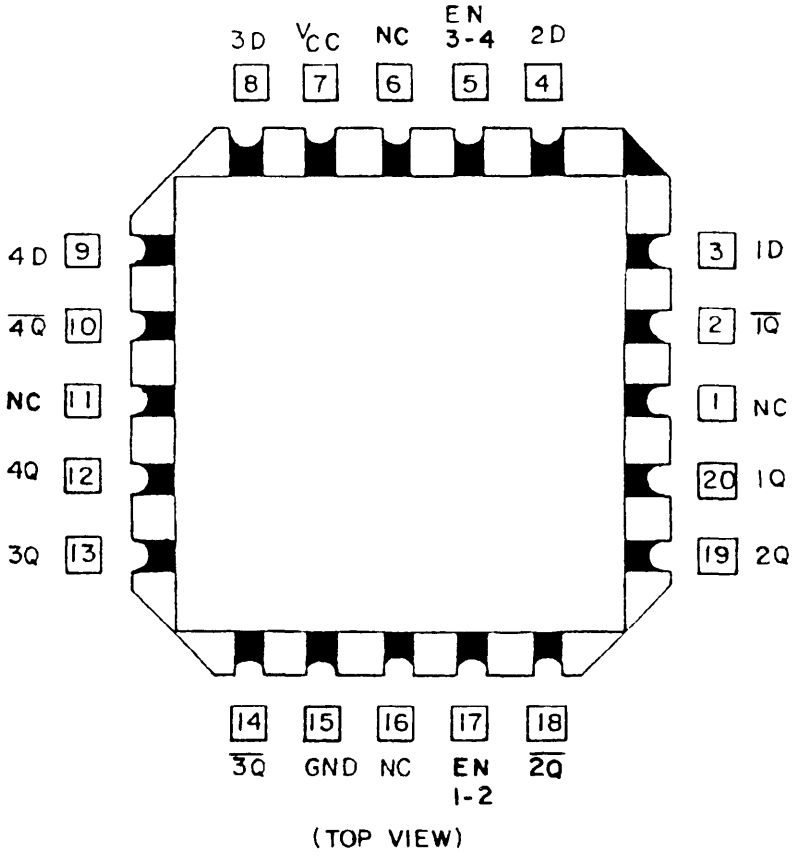
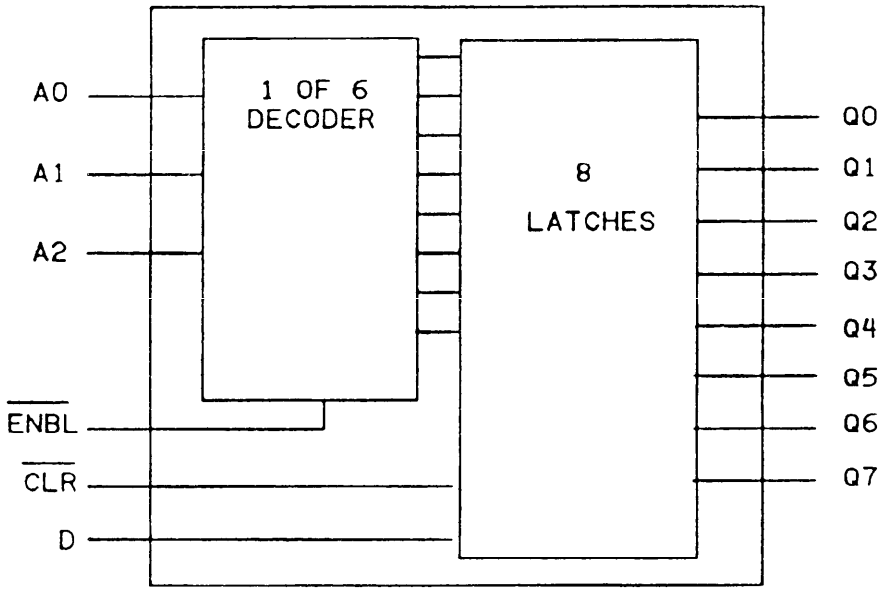


FIGURE 1. Logic diagrams and terminals connections - Continued.

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Device type 02

Cases E and F

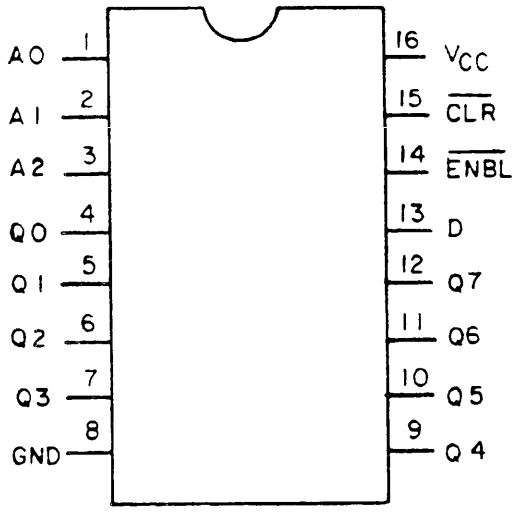


FIGURE 1. Logic diagrams and terminals connections - Continued.

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Device Type 02

Case 2

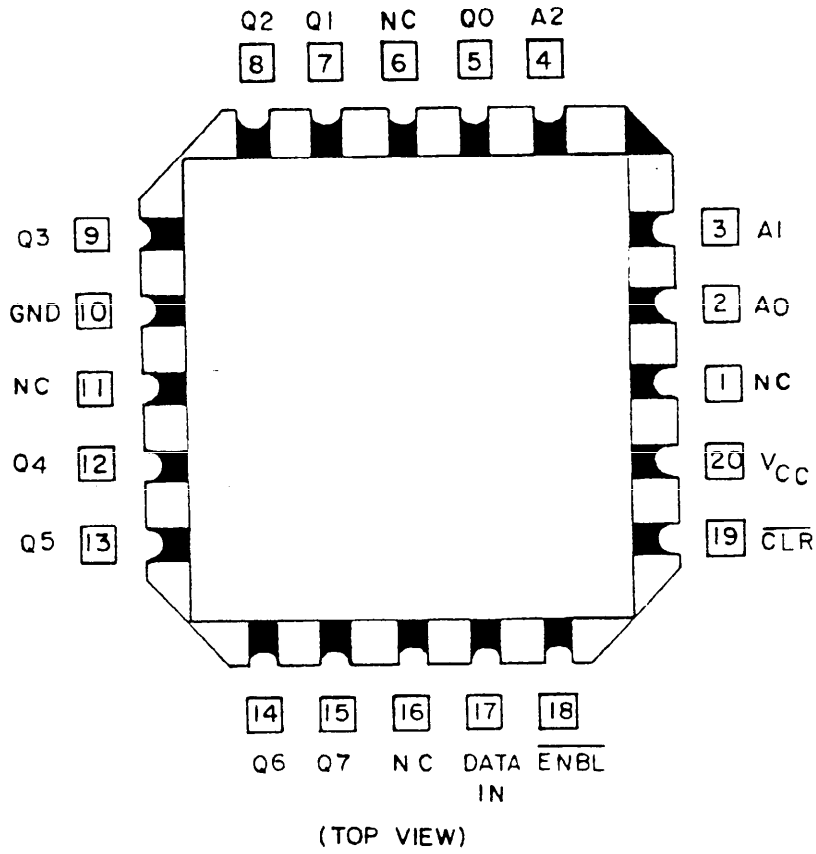
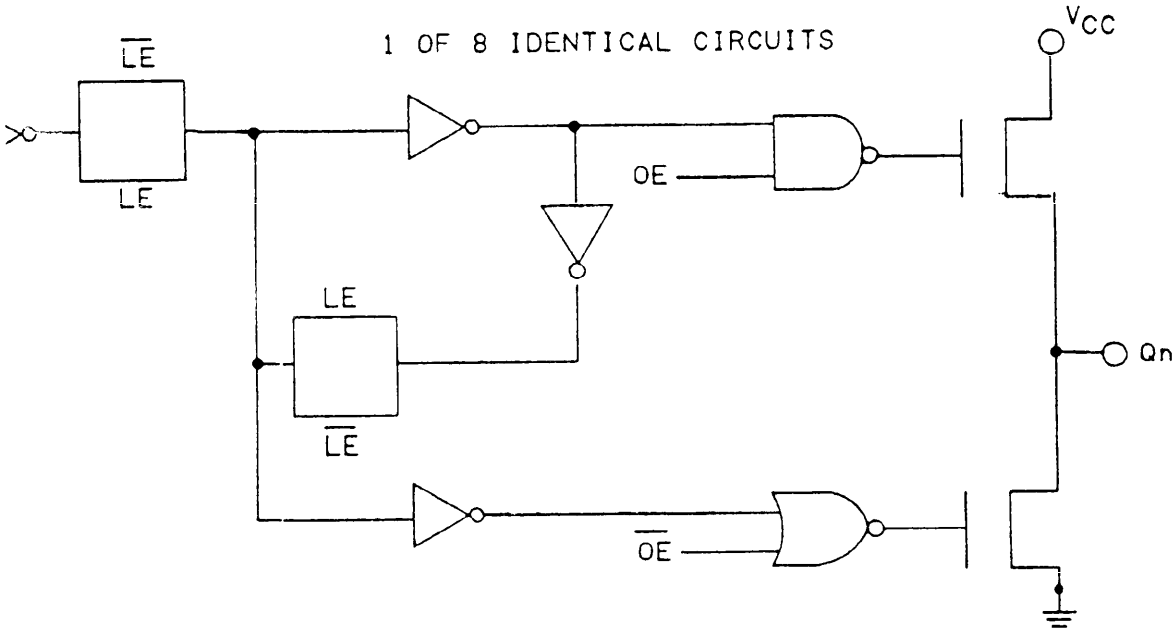


FIGURE 1. Logic diagrams and terminals connections - Continued.

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Devices types 03 and 53



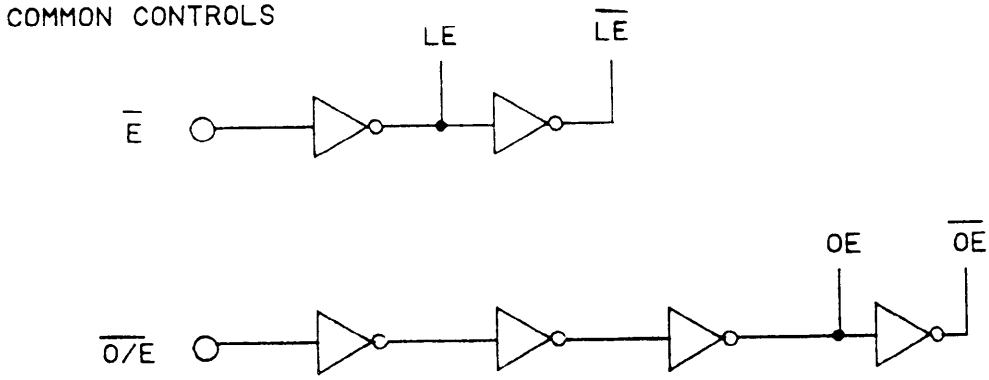
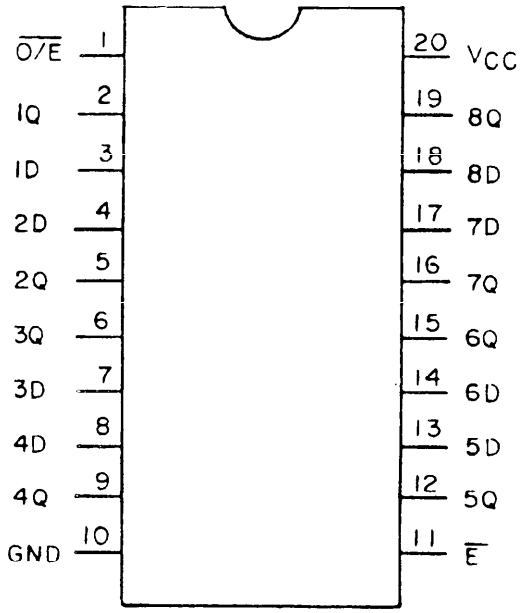


FIGURE 1. Logic diagrams and terminals connections - Continued.

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Device type 03 and 53

Cases R and S



Device type 03 and 53

Case 2

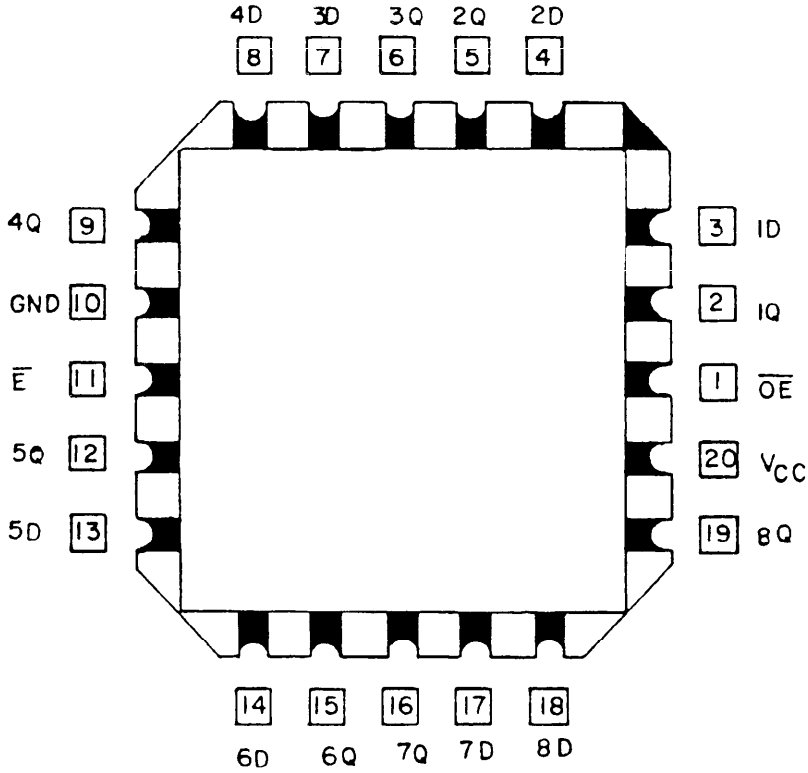


FIGURE 1. Logic diagrams and terminals connections - Continued.

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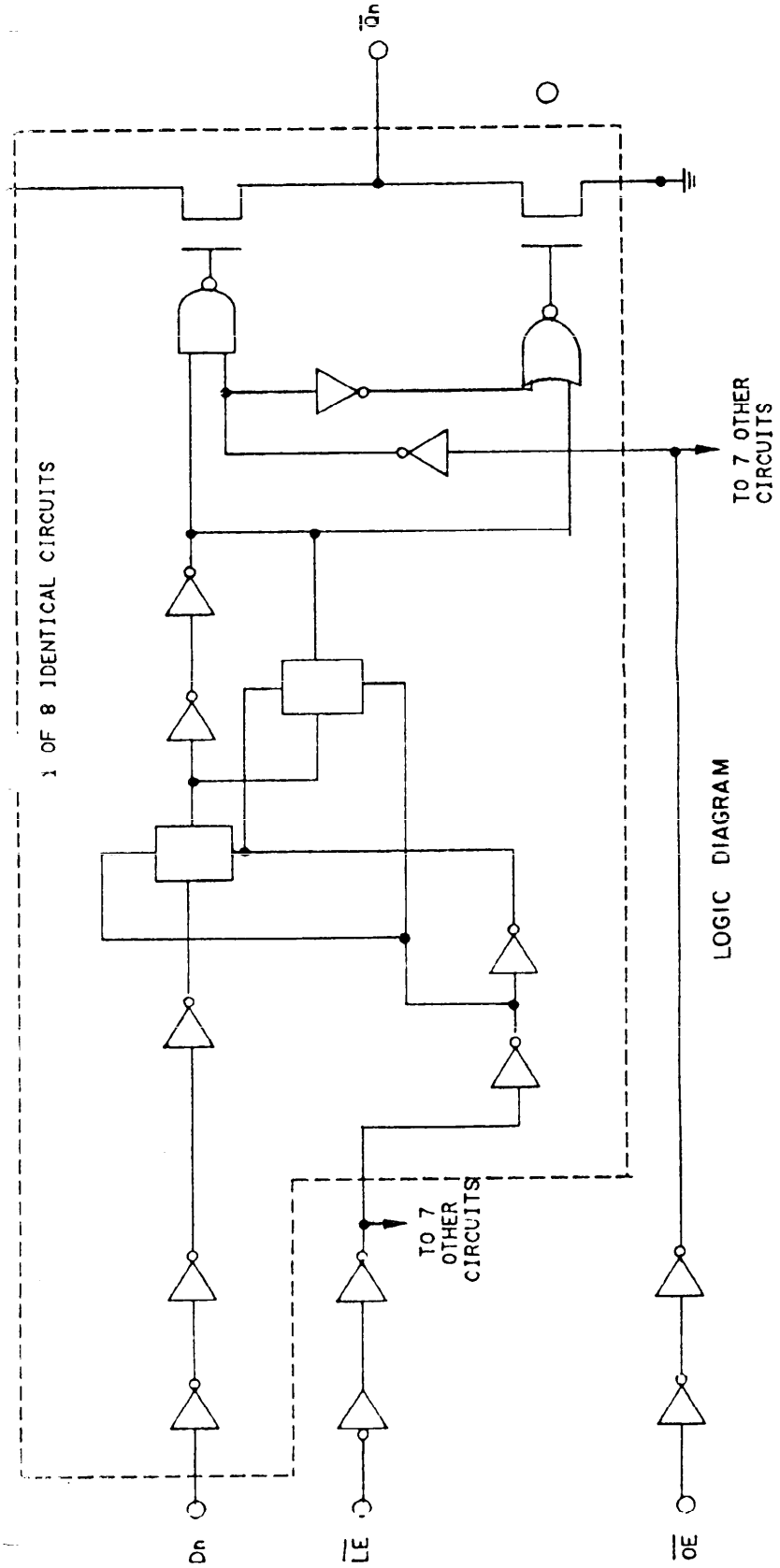
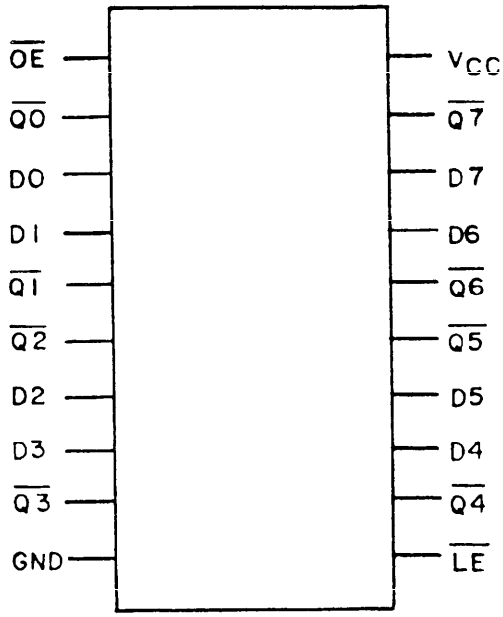


FIGURE 1. Logic diagrams and terminals connections - Continued.

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Device type 04
Cases R and S



Device type 04
Case 2

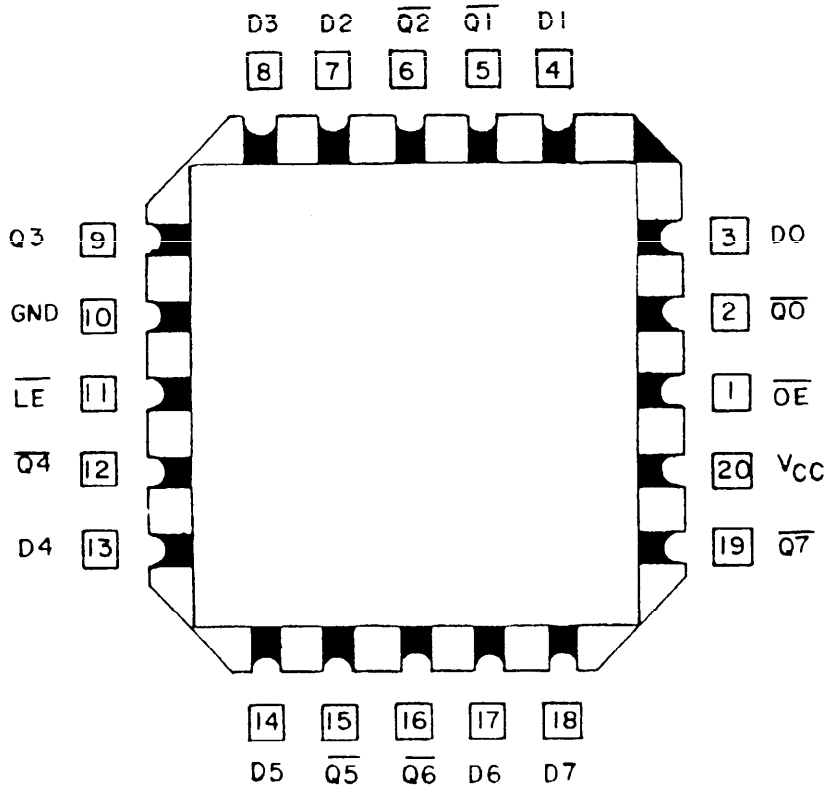


FIGURE 1. Logic diagrams and terminals connections - Continued.

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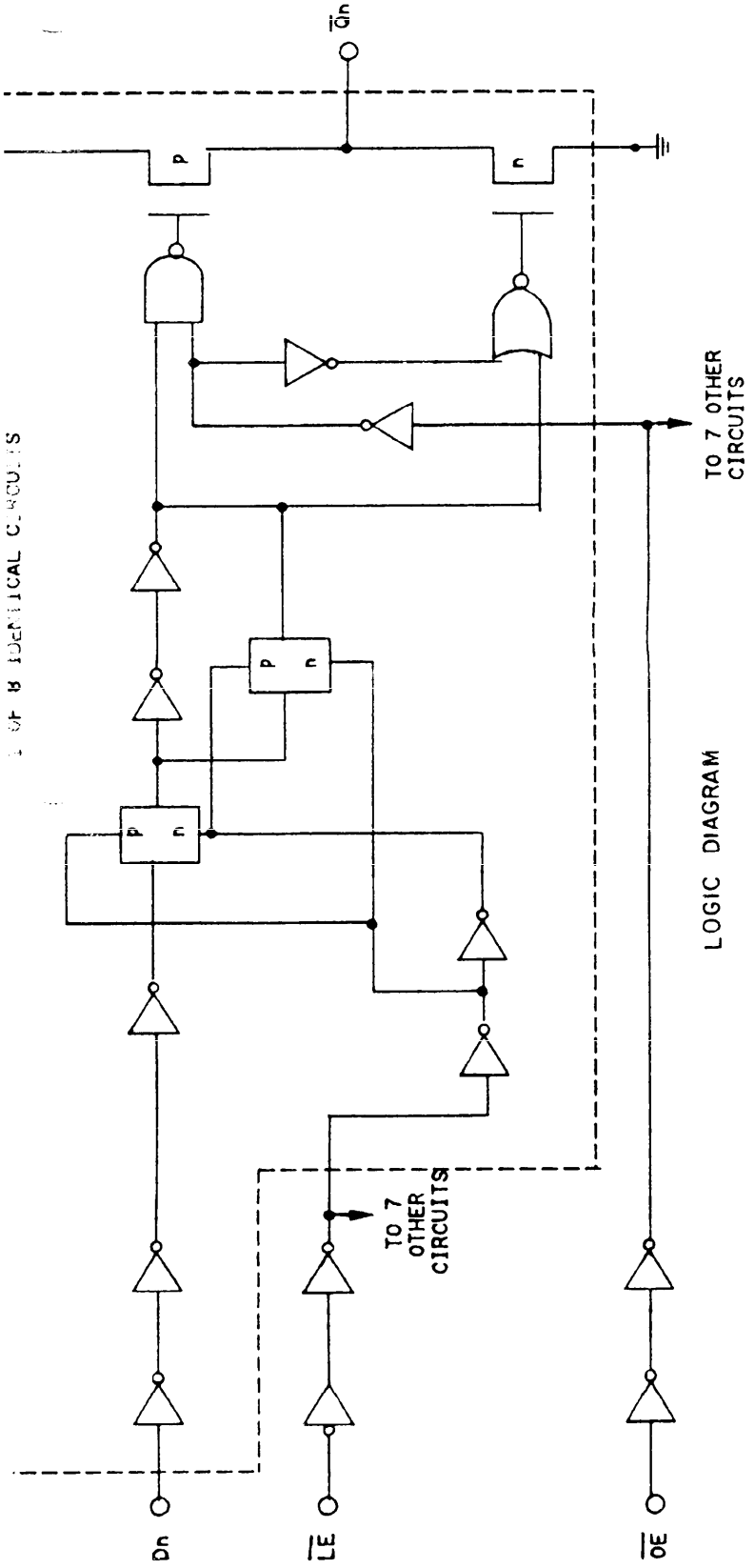
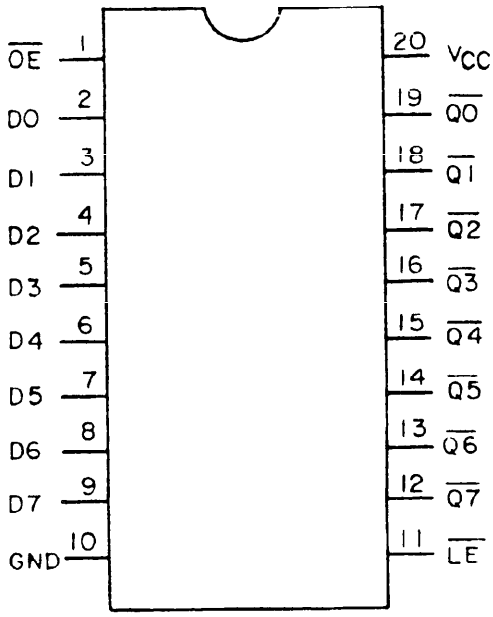


FIGURE 4. Switching time test circuit - Continued.

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Device type 05

Cases R and S



Device type 05

Case 2

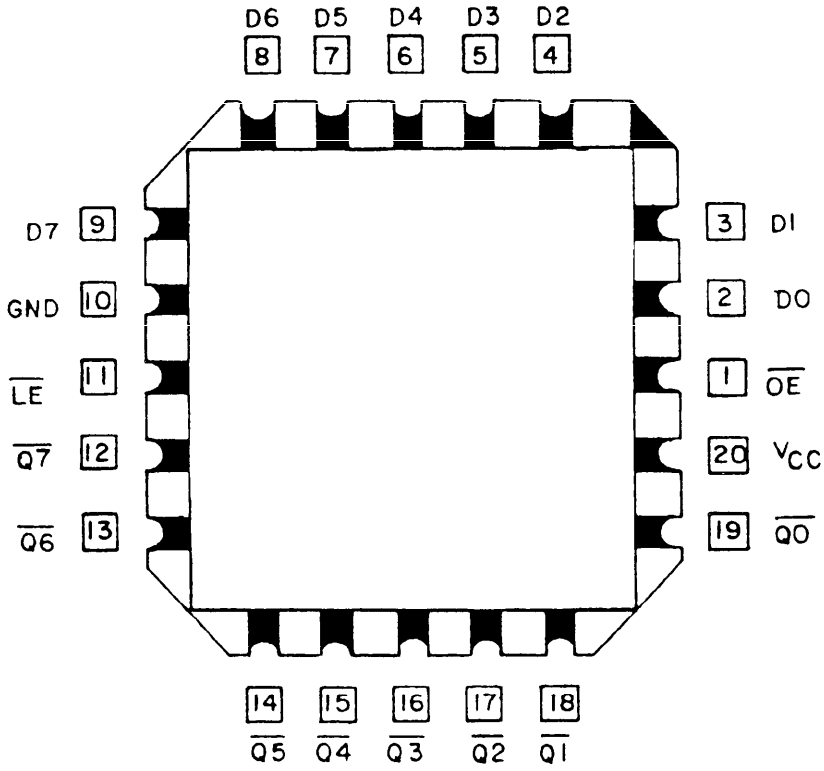
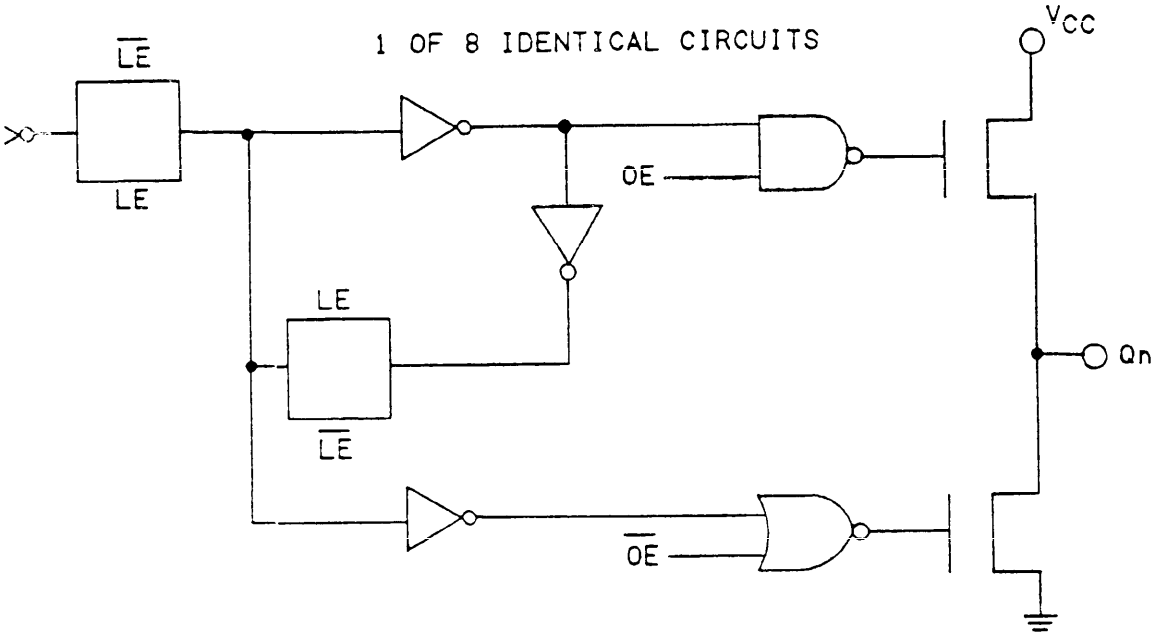


FIGURE 1. Logic diagrams and terminals connections - Continued.

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Device type 08



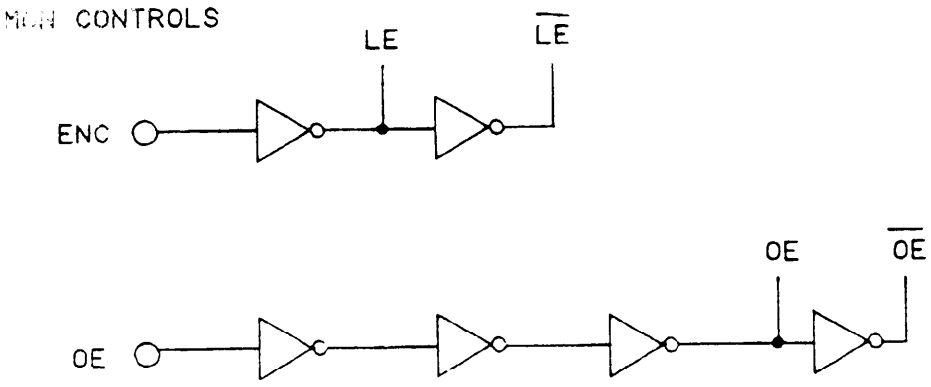
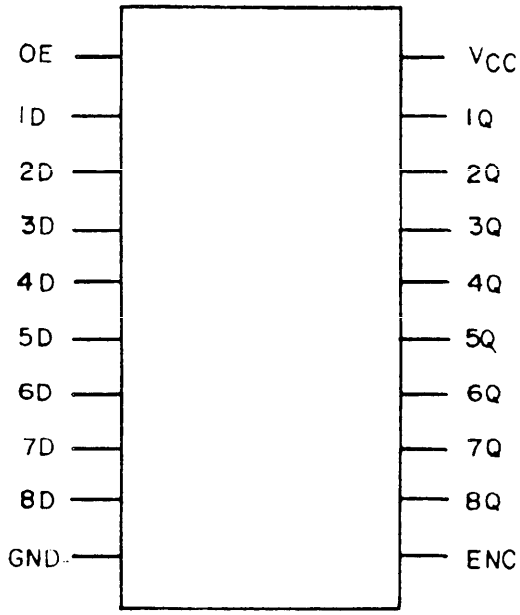


FIGURE 1. Logic diagrams and terminals connections - Continued.

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Device type 06

Cases R and S



Device type 06

Case 2

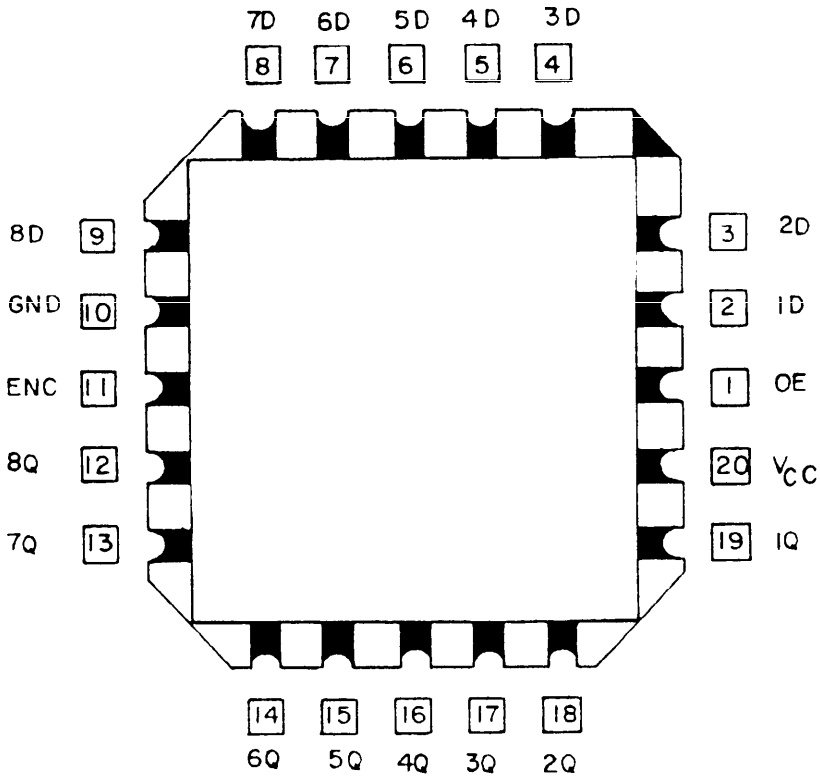


FIGURE 1. Logic diagrams and terminals connections - Continued.

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Device type 01

Inputs		Outputs	
D	$\overline{EN}$	Q	$\overline{Q}$
L	H	L	H
H	H	H	L
X	L	Qo	$\overline{Q}$o

H = High Level
L = Low Level
X = Don't care
Qo = The level of Q before the transition of $\overline{EN}$.

Device types 02

Inputs		Output of address latch	Each other Output	Function
$\overline{CLR}$	$\overline{EN}$			
H	L	D	Qio	Addresable latch
H	H	Qio	Qio	Memory
L	L	D	L	8-line Demultiplexer
L	H	L	L	Reset

H = High Level
L = Low Level
D = The level at the data input
Qio = The level of Q (i=0, 1 7), as appropriate) before the indicated steady-state input conditions were established.

Latch Selection Table

Select Inputs			Latch Addressed
A_2	A_1	A_0	
L	L	L	0
L	L	H	1
L	H	L	2
L	H	H	3
H	L	L	4
H	L	H	5
H	H	L	6
H	H	H	7

FIGURE 2. Truth tables.

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Device type 03 and 53

$\overline{O/E}$	$\overline{E}$	D Data	Q Output
L	H	H	H
L	H	L	L
L	L	1	L
L	L	h	H
H	X	X	Z

NOTE:
L = Low voltage level
H = High voltage level
1 = Low voltage level one setup time prior to the high to low latch enable transition
h = High voltage level one setup time prior to the high to low latch enable transition
X = Don't Care
Z = High impedance state

Output Enable $\overline{OE}$	Latch Enable $\overline{LE}$	D Data	Q Output
L	H	H	L
L	H	L	h
L	L	1	H
L	L	h	L
H	X	X	Z

NOTE:
L = Low voltage level.
H = High voltage level.
1 = Low voltage level one set-up time prior to the high to low latch enable transition.
h = High voltage level one set-up time prior to the high to low latch enable transition.
X = Don't care.
Z = High impedance state.

Device type 05

Output Enable $\overline{OE}$	Latch Enable $\overline{LE}$	D Data	Q Output
L	H	H	L
L	H	L	H
L	L	1	H
L	L	h	L
H	X	X	Z

NOTE:
L = Low voltage level
H = High voltage level
1 = Low voltage level one setup time prior to the high to low latch enable transition
h = High voltage level one setup time prior to the high to low latch enable transition
X = Don't Care
Z = High impedance state

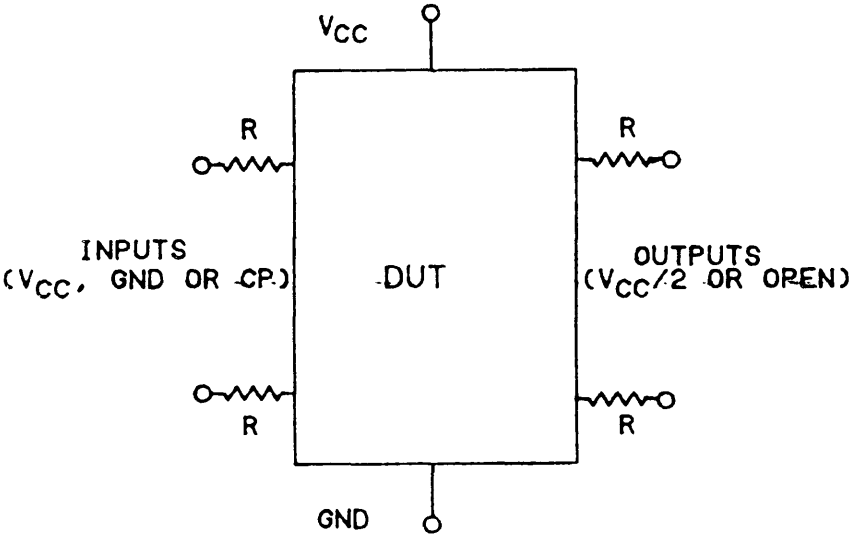
Device Type 06

$\overline{O/E}$	$\overline{ENC}$	D Data	Q Output
L	H	H	H
L	H	L	L
L	L	1	L
L	L	h	H
H	X	X	Z

NOTE:
L = Low voltage level
H = High voltage level
1 = Low voltage level one setup time prior to the high to low latch enable transition
h = High voltage level one setup time prior to the high to low latch enable transition
X = Don't Care
Z = High impedance state

FIGURE 2. Truth tables - Continued.

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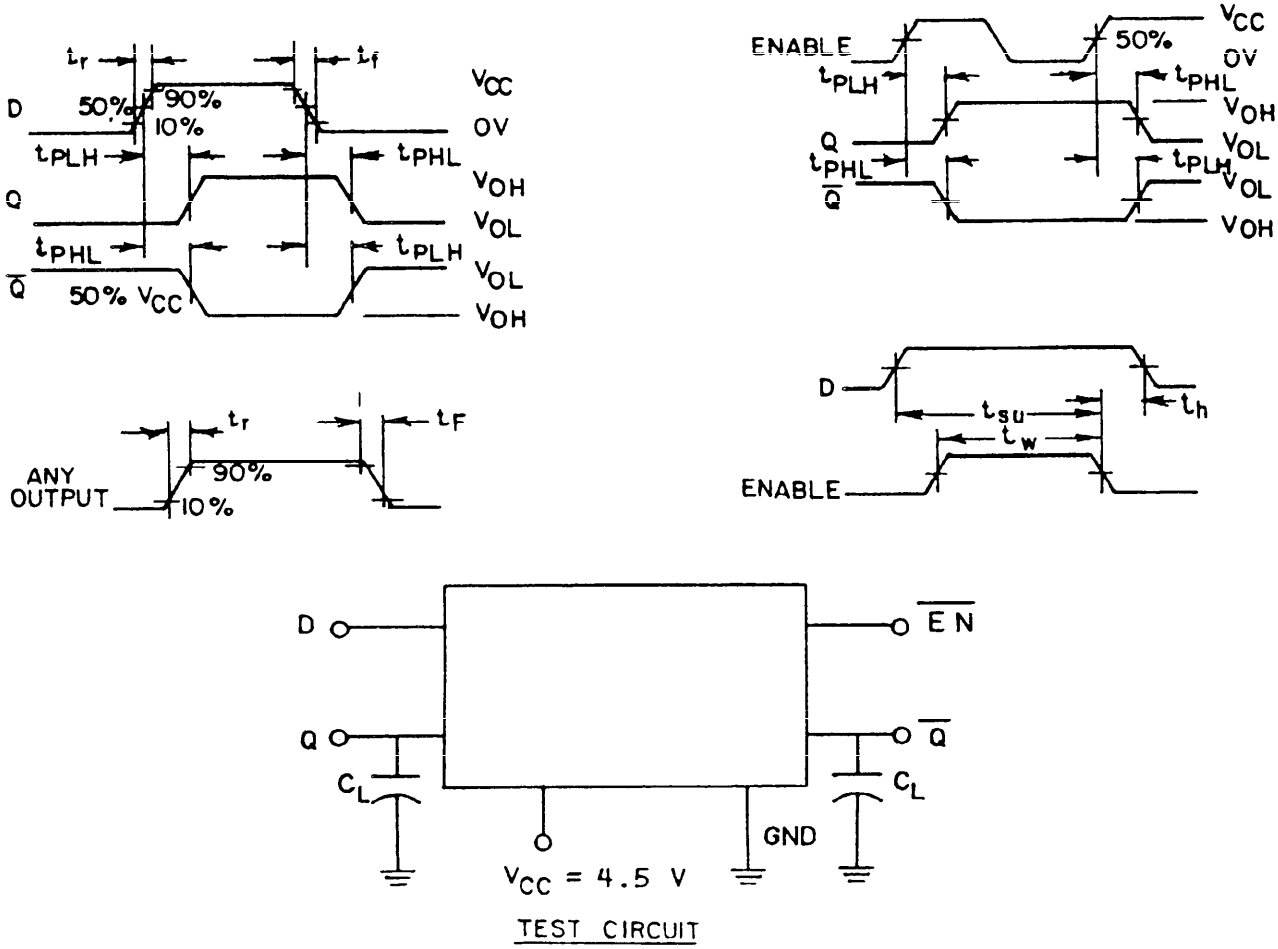


NOTES:
1. For static burn-in I, all inputs shall be connected to GND. Outputs shall be open or connected to $V_{CC}/2$. Resistors are optional on outputs if open. Resistors are required on inputs and outputs connected to $V_{CC}/2$. R = 680 Ω to 47 kΩ.
2. For static burn-in II, all inputs shall be connected through a resistor to V_{CC}. Outputs shall be open or connected to $V_{CC}/2$. Resistors are optional on outputs if open. Resistors are required on inputs and on outputs connected to $V_{CC}/2$. R = 680 to 47 kΩ.
3. For dynamic burn-in, all inputs shall be connected through the resistors in parallel to a common CP. Outuputs shall be connected to $V_{CC}/2$ ±0.5 V through the resistors. R = 1 kΩ for outputs for device types 01, 02, and 680Ω ±5% for device types 03 - 06, and 53. R = 1 kΩ to 47 kΩ for inputs of device types 01, 02, and 680Ω to 47 kΩ for inputs of device types 03 - 06, and 53.
4. CP = 25 kHz to 1 MHz square wave; duty cycle = 50 ±15%; V_{IH} = 4.5 V to V_{CC}; V_{IL} = 0 ±0.5 V transition time ≤ 0.5 µs.
5. V_{CC} = 6.0 V ±0.5 V, V_{CC} = 5.5 V ±0.5 V for device 53.

FIGURE 3. Burn-in and life test circuits.

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Device type 01



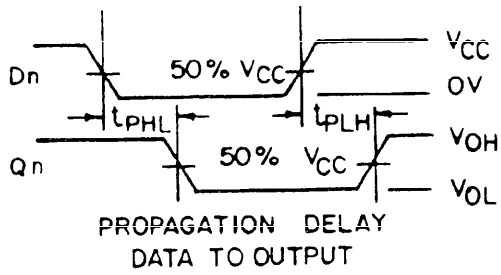
TEST CIRCUIT

NOTES:
1. Clock input pulse characteristics: $t_R = t_F \leq 6$ ns.
2. Data or enable input pulse characteristics: $t_R = t_F = 6$ ns; $t_{SETUP} \leq 30$ ns; $t_{HOLD} = 8$ ns and $t_{PULSE\ WIDTH} = 24$ ns.
3. $C_L = 50$ pF.
4. Voltage measurements are to be made with respect to network ground terminal.

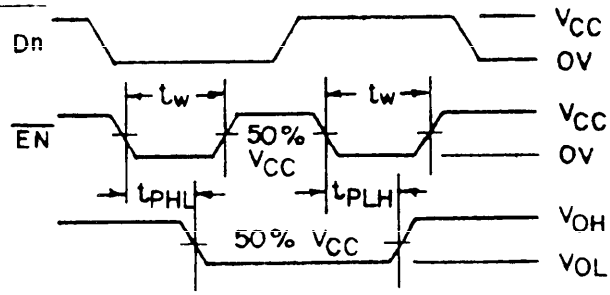
FIGURE 4: Switching time test circuit.

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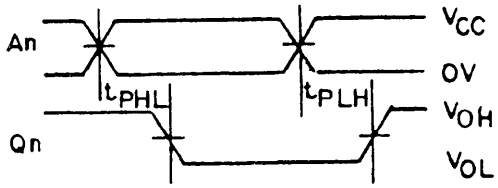
Device type 02



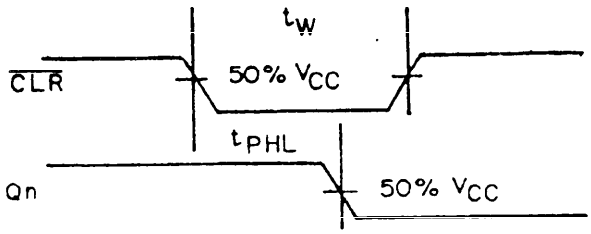
PROPAGATION DELAY
DATA TO OUTPUT



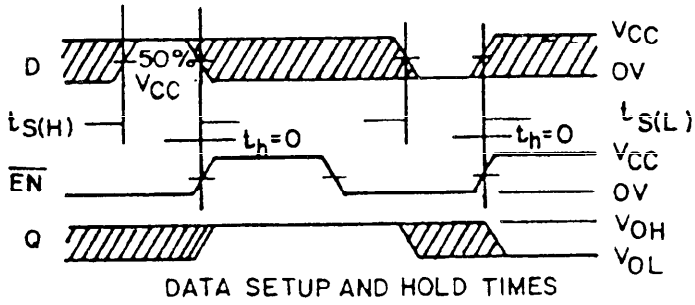
LATCH ENABLE PULSE WIDTH AND
LATCH ENABLE TO OUTPUT
PROPAGATION DELAY



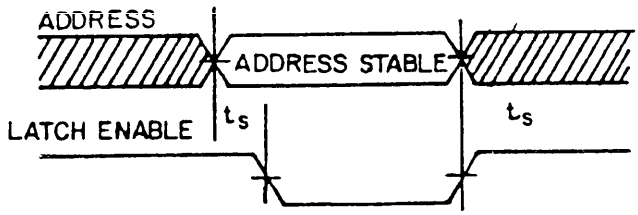
PROPAGATION DELAY
ADDRESS TO OUTPUT



$\overline{CLR}$ TO OUTPUT DELAY
AND $\overline{CLR}$ PULSE WIDTH



DATA SETUP AND HOLD TIMES

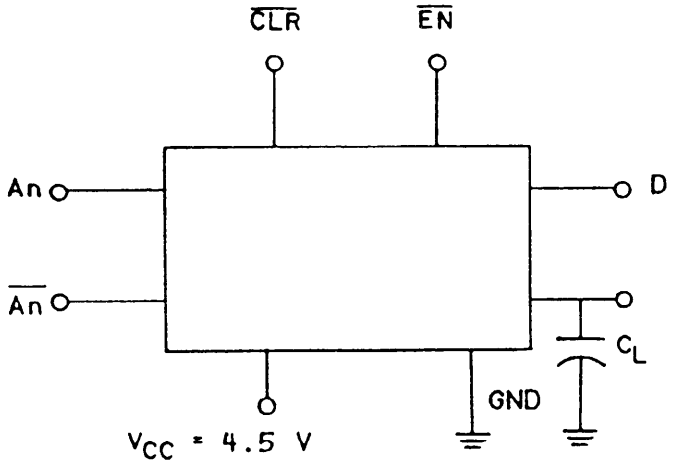


ADDRESS SETUP AND HOLD TIMES

AC WAVEFORMS

FIGURE 4. Switching time test circuit - Continued.

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TEST CIRCUIT

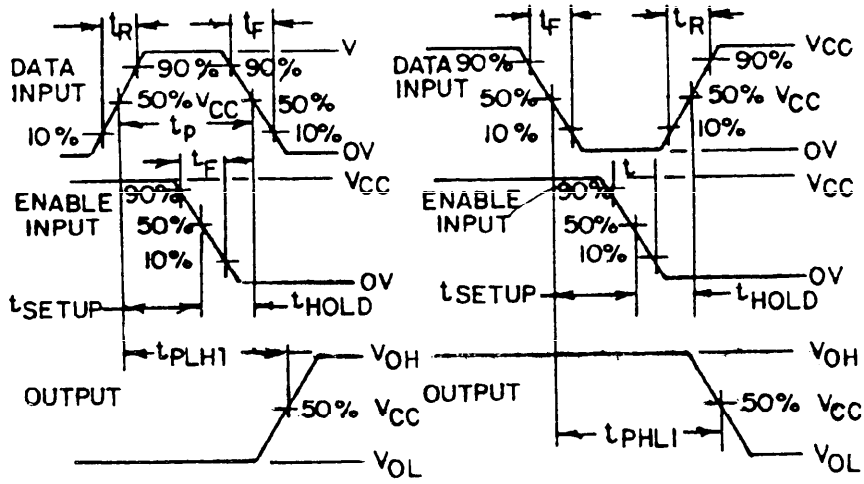
Device type 02

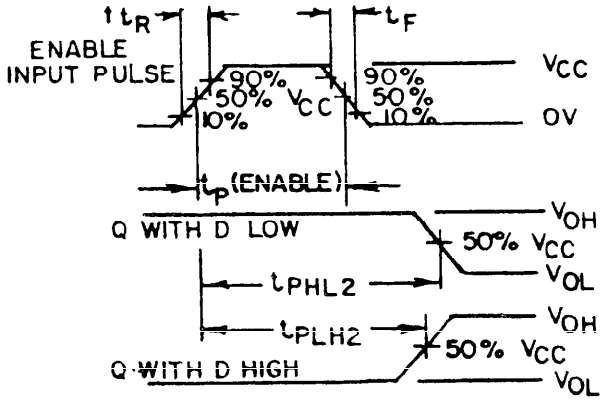
NOTES:
1. Clock input pulse characteristics: $t_R = t_F \leq 6$ ns.
2. Data, address, latch enable or clear input pulse characteristics: $t_R = t_F = 6$ ns; $t_{SETUP} \leq 30$ ns; $t_{PULSE\ WIDTH} \leq 24$ ns and $t_{HOLD} \leq 24$ ns.
3. $C_L = 50$ pF.
4. Voltage measurements are to be made with respect to network ground terminal.

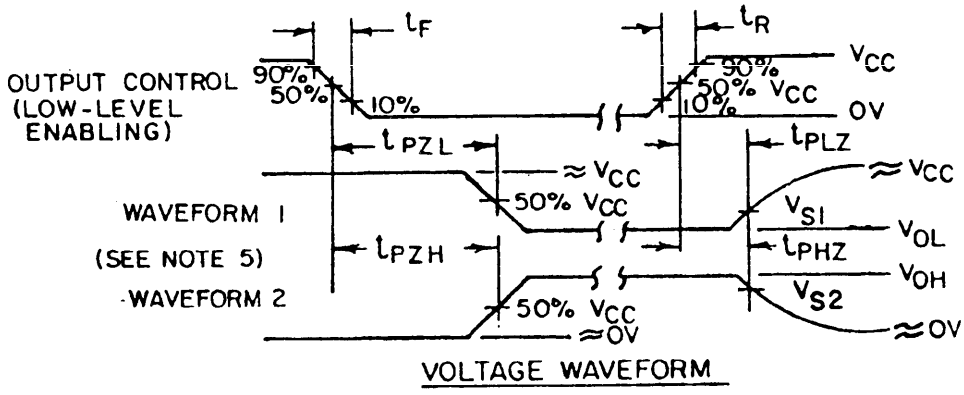
FIGURE 4: Switching time test circuit - Continued.

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Device types 03 and 53.



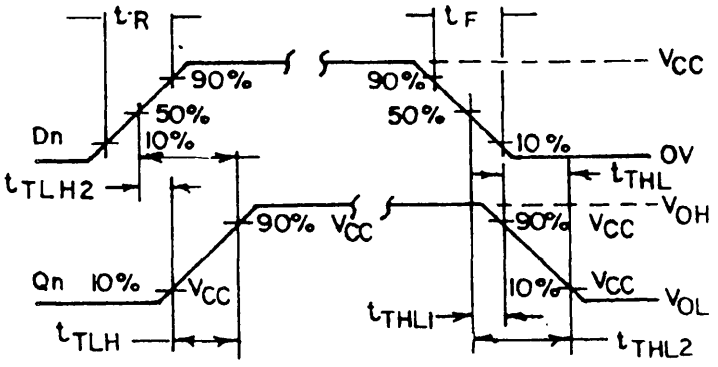




VOLTAGE WAVEFORM

FIGURE 4. Switching Time Test Circuit - Continued.

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$t_{TLH} = t_{TLH1} - t_{TLH2}$ $t_{THL} = t_{THL2} - t_{THL1}$

Device types 03 and 53.

NOTES:
1. Clock input pulse characteristics: $t_R = t_F \leq 6$ ns.
2. Data, latch enable and output enable input pulse characteristics: $t_R = t_F \leq 6$ ns; $t_{SETUP} \geq 30$ ns; $t_{PULSE\ WIDTH} \geq 30$ ns and $t_{HOLD} \geq 15$ ns.
3. $C_L = 50$ pF and $R_L = 1000$ ohms.
4. Voltage measurements are to be made with respect to network ground terminal.
5. For t_{PHZ} and t_{PZH}, a 1 kΩ resistor is connected between the output and GND terminal. For t_{PZL} and t_{PLZ}, a 1 kΩ resistor is connected between the output and V_{CC} terminal. $V_{S1} = V_{OL} + 0.1\ V\ (V_{OH} - V_{OL})$. $V_{S2} = V_{OH} - 0.1\ V\ (V_{OH} - V_{OL})$.
6. The input signal for HCT device (type 53) will be 0 - 3 volts; however, the 50% V_{CC} measure point is 1.3 volts for input and output signals. The 10% V_{CC} and 90% V_{CC} points are 0.3 volts and 2.7 volts, respectively.

FIGURE 4. Switching time test circuit - Continued.

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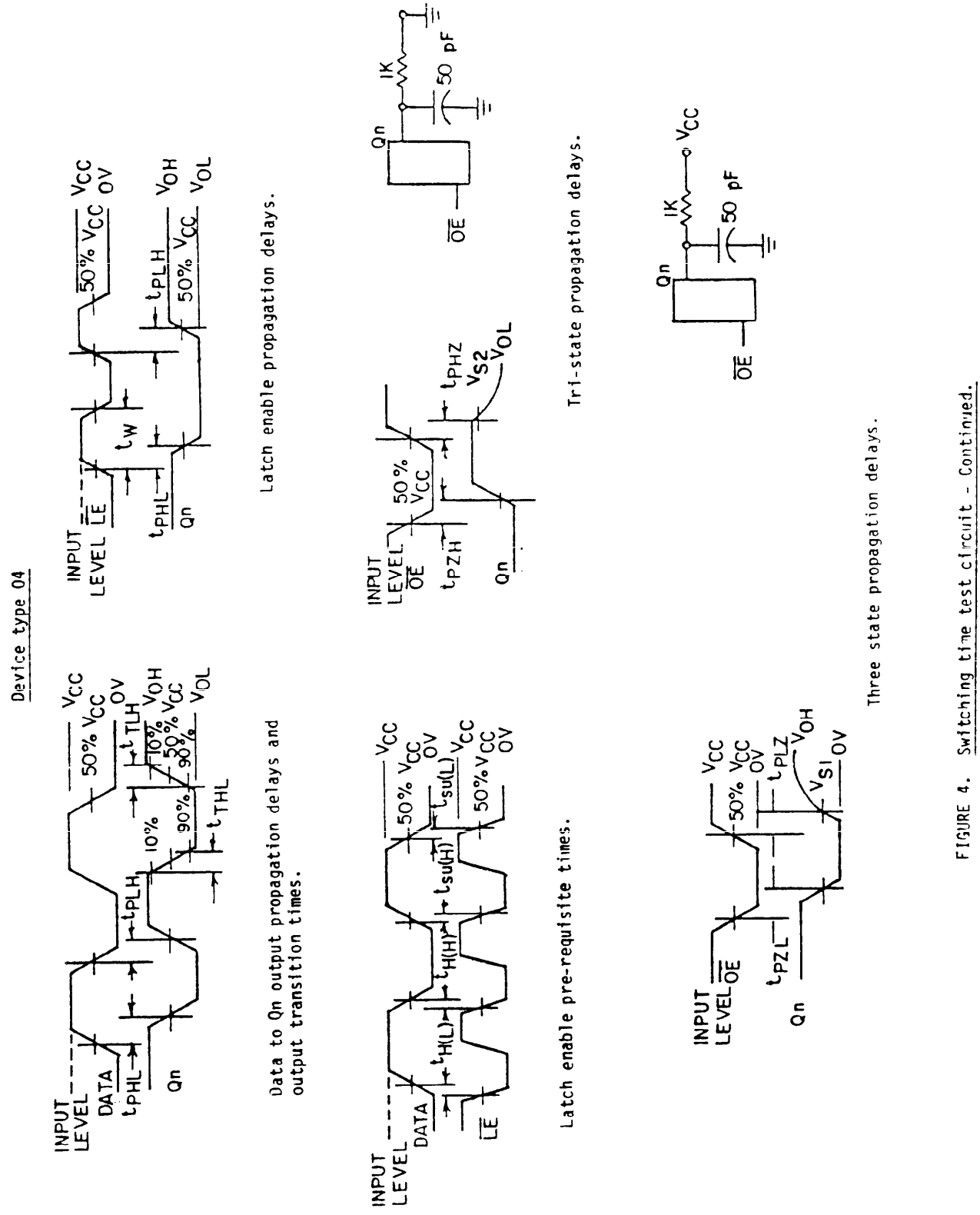
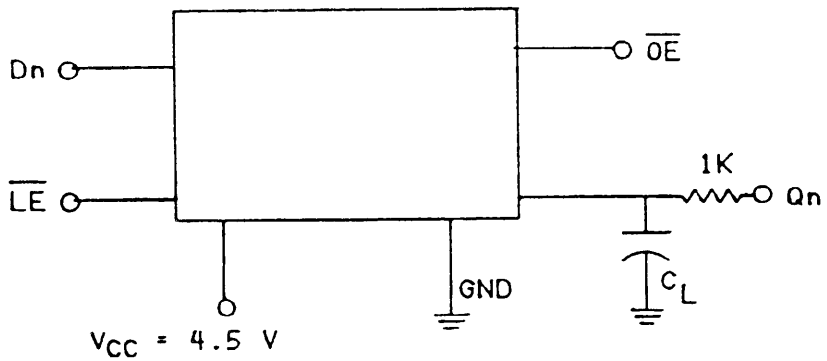


FIGURE 4. Switching time test circuit - Continued.

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TEST CIRCUIT

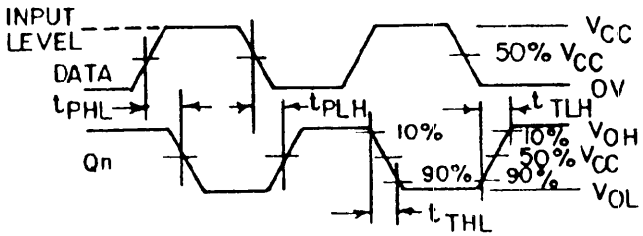
Device type 04

NOTES:
1. Clock input pulse characteristics: $t_R = t_F \leq 6$ ns.
2. Data, latch enable and output enable input pulse characteristics: $t_R = t_F \leq 6$ ns; $t_{SETUP} \leq 30$ ns; $t_{PULSE\ WIDTH} \leq 30$ ns and $t_{HOLD} \leq 15$ ns.
3. $C_L = 50$ pF and $R_L = 1000$ ohms.
4. Voltage measurements are to be made with respect to network ground terminal.
5. For t_{PHZ} and t_{PZH}, a 1 kΩ resistor is connected between the output and GND terminal. For t_{PZL} and t_{PLZ}, a 1 kΩ resistor is connected between the output and V_{CC} terminal. $V_{S1} = V_{OL} + 0.1$ V ($V_{OH} - V_{OL}$). $V_{S2} = V_{OH} - 0.1$ V ($V_{OH} - V_{OL}$).

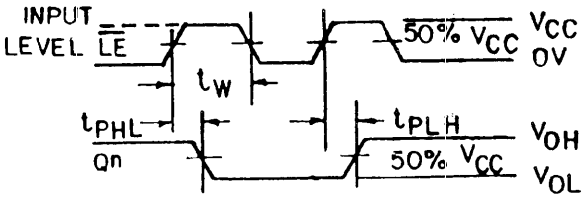
FIGURE 4. Switching time test circuit - Continued.

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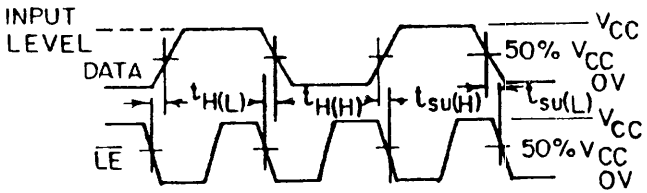
document to the extent specified herein. Unless otherwise specified, the issues of these documents are



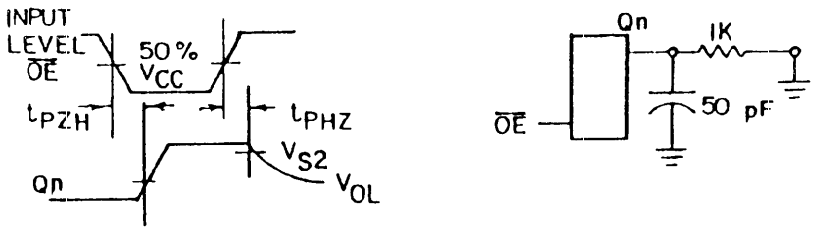
Data to Qn output propagation delays and output transition times.



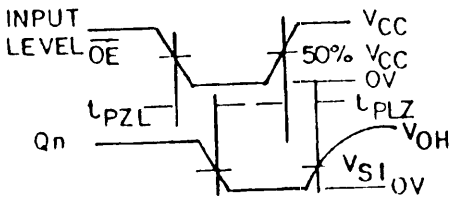
Latch enable propagation delays.

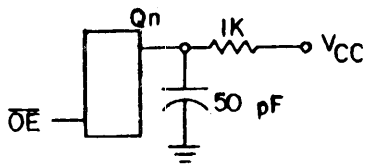


Latch enable pre-requisite times.



Tri-state propagation delays.

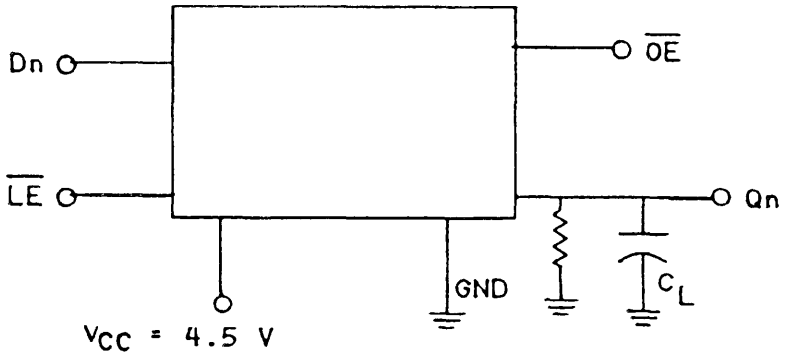




Three state propagation delays.

FIGURE 4. Switching time test circuit - Continued.

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TEST CIRCUIT

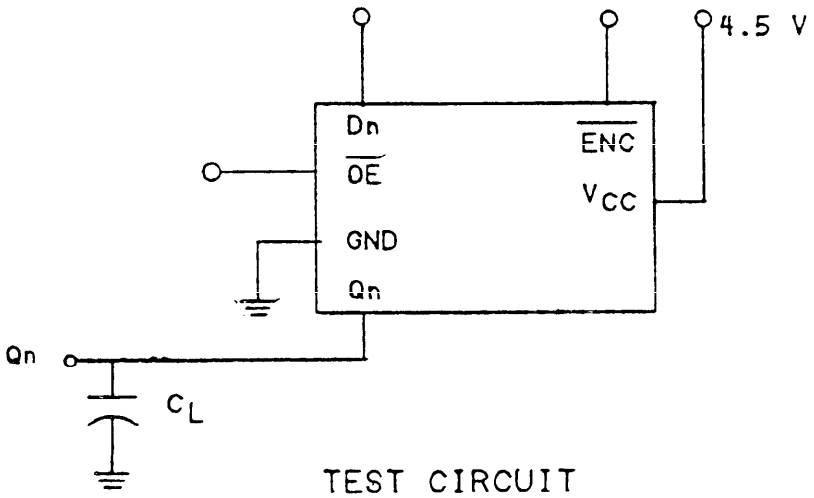
Device type 05

NOTES:
1. Clock input pulse characteristics: $t_R = t_F \leq 6$ ns.
2. Data, latch enable and output enable input pulse characteristics: $t_R = t_F \leq 6$ ns; $t_{SETUP} \leq 23$ ns; $t_{PULSE\ WIDTH} \leq 24$ ns and $t_{HOLD} \leq 15$ ns.
3. $C_L = 50$ pF and $R_L = 1000$ ohms.
4. Voltage measurements are to be made with respect to network ground terminal.
5. For t_{PHZ} and t_{PZH}, a 1 kΩ resistor is connected between the output and GND terminal. For t_{PZL} and t_{PLZ}, a 1 kΩ resistor is connected between the output and V_{CC} terminal. $V_{S1} = V_{OL} + 0.1$ V ($V_{OH} - V_{OL}$). $V_{S2} = V_{OH} - 0.1$ V ($V_{OH} - V_{OL}$).
6. All trip points are referenced to V_{CC}.

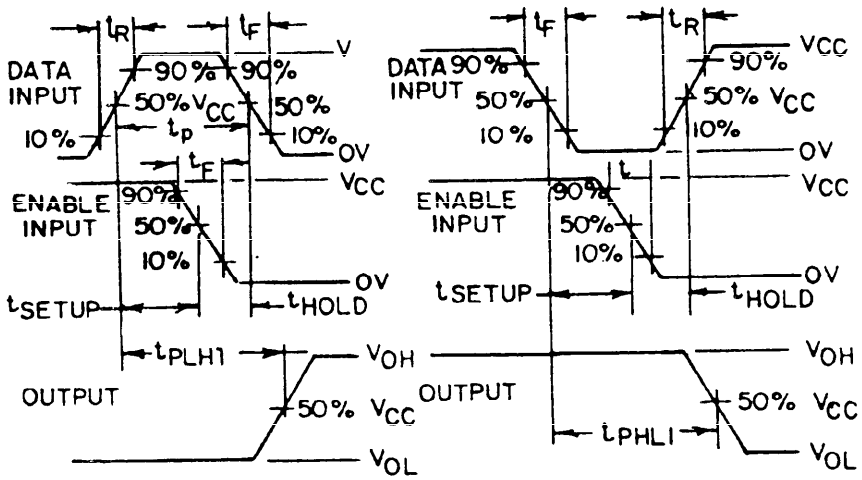
FIGURE 4. Switching time test circuit - Continued.

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Device type 06



TEST CIRCUIT



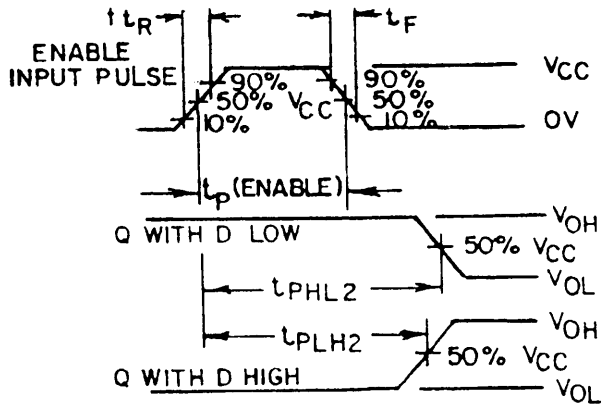
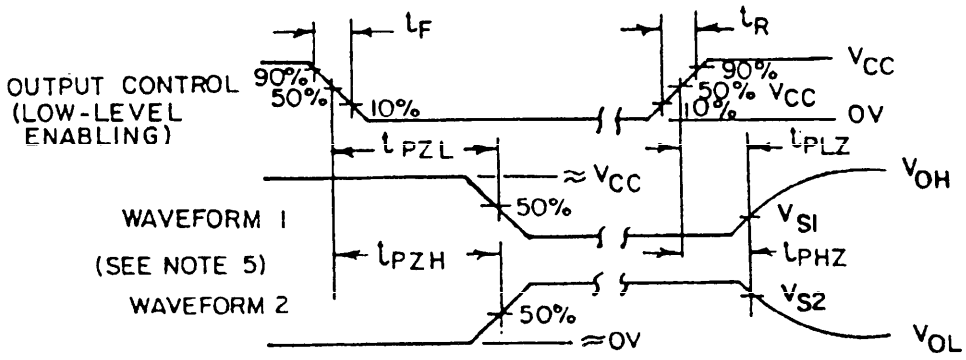


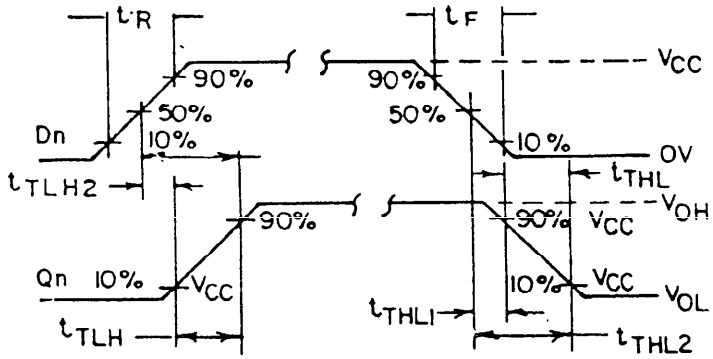
FIGURE 4. Switching Time Test Circuit - Continued.

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Device Type 06



VOLTAGE WAVEFORM



$t_{TLH} = t_{TLH1} - t_{TLH2}$ $t_{THL} = t_{THL2} - t_{THL1}$

Device Type 06

NOTES:
1. Clock input pulse characteristics: $t_R = t_F \leq 6$ ns.
2. Data, latch enable and output enable input pulse characteristics: $t_R = t_F \leq 6$ ns; $t_{SETUP} \leq 23$ ns; $t_{PULSE\ WIDTH} \leq 24$ ns and $t_{HOLD} \leq 15$ ns.
3. $C_L = 50$ pF and $R_L = 1000$ ohms.
4. Voltage measurements are to be made with respect to network ground terminal.
5. For t_{PHZ} and t_{PZH}, a 1 kΩ resistor is connected between the output and GND terminal. For t_{PZL} and t_{PLZ}, a 1 kΩ resistor is connected between the output and V_{CC} terminal. $V_{S1} = V_{OL} + 0.1\ V\ (V_{OH} - V_{OL})$. $V_{S2} = V_{OH} - 0.1\ V\ (V_{OH} - V_{OL})$.
6. All trip points are referenced to V_{CC}.

FIGURE 4. Switching time test circuit - Continued.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Case 2 / Case E	Terminal conditions: 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		Test no.	1/	1Q	2Q	EN3-4	V_{CC}	3Q	4Q	4/	4Q	3Q	3/	GND	EN1-2	2Q	2J	1Q												
V_{IC} (pos)		1		1 mA	1 mA	1 mA	GND	1 mA	1 mA					1/	1 mA								1Q 2Q EN3 4 3Q 4Q EN1-2		1.5					V
		2					"							"																
		3					"							"																
		4					"							"																
		5					"							"																
		6					"							"																
V_{IC} (neg)		7		-1 mA	-1 mA	-1 mA	1/	-1 mA	-1 mA					1/	-1 mA								1Q 2Q EN3-4 3Q 4Q EN1-2		-1.5					V
		8					"							"																
		9					"							"																
		10					"							"																
		11					"							"																
		12					"							"																
I_{CC}	3005	13		6.0 V GND	6.0 V GND	6.0 V GND	6.0 V	6.0 V GND	6.0 V GND					"	6.0 V GND								V_{CC} V_{CC}		.1 .1		10 10			μA μA
		14																												
V_{OH3}	3006	15		4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	"			-20 μA		"	4.2 V		-20 μA	-20 μA	4.2 V			-20 μA	1Q 2Q 3Q 4Q 1Q 2Q 3Q 4Q	5.95		5.95		5.95		V
		16																												
		17																												
		18	-20 μA	1.2 V	1.2 V	1.2 V	4.2 V	1.2 V	4.2 V		-20 μA				4.2 V				4.2 V		-20 μA									
		19					4.2 V																							
		20																												
		21																												
		22					4.2 V				1.2 V		-20 μA																	
V_{OH5}	3006	23-30	Same tests and terminal conditions as V_{OH3} except I_{OH} = -5.2 mA.																					5.48		5.2		5.48		V
V_{OL3}	3007	31		1.2 V	1.2 V			1.2 V						GND	4.2 V				4.2 V			20 μA	1Q 2Q 3Q 4Q 1Q 2Q 3Q 4Q		.05		.05		.05	V
		32												"																
		33												"																
		34												"																
		35	20 μA	4.2 V		1.2 V	4.2 V		6.0 V	1.2 V	1.2 V		20 μA	"	4.2 V	20 μA	20 μA	20 μA		20 μA										
		36			4.2 V			4.2 V						"					4.2 V											
		37					4.2 V			4.2 V				"																
		38					4.2 V				4.2 V	20 μA		"																
V_{OH5}	3006	39-46	Same tests and terminal conditions as V_{OL3} except I_{OH} = 5.2 mA.																						.20		.4		.26	V
I_{OS4}	3011	47		4.0 V	4.0 V	4.0 V	4.0 V	4.0 V	4.0 V					GND	4.0 V				4.0 V			GND	1Q 2Q 3Q 4Q 1Q 2Q 3Q 4Q	-10	-120	-10	-120	-10	-120	mA
		48												"																
		49												"																
		50												"																
		51	GND	GND	GND	GND	4.0 V	4.0 V	"	4.0 V	4.0 V	GND	GND	"	4.0 V	GND	GND	GND	4.0 V	GND	GND	GND									
		52												"																
		53												"																
		54					4.0 V				4.0 V			"																
I_{IH}	3010	55		6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V					"	6.0 V								1Q 2Q EN3-4 3Q 4Q EN1-2		.05		.1			μA
		56												"																
		57												"																
		58												"																
		59												"																
		60												"																

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 test method	Case outline	Terminal conditions																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
I_{IL}	3009		20	GND		EN3-4		GND	V_{CC}						EN1-2				10 20 EN3-4 30 40 EN1-2		-0.35 " " " "		-1 " " " "			μA " " " "
I_{CCH}	3002			2/				GND	" " " 2/						2/				10 20 30 40 EN3-4 EN1-2	Subgroup 4 T_C = +25°C	10 " " " " "					mA " " " " "
																				Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 9 T_C = -55°C		
Truth table 2/	3014				A A B B	A A B B	4.5 V " " "	A A B B	4.5 V " " "			L H H L	L H H L	GND " " "	A B B A				All outputs 4/ "							
																				Subgroup 10 T_C = +25°C		Subgroup 11 T_C = +125°C		Subgroup 12 T_C = -55°C		
t_{PLH1}	3003 (Fig. 4)			IN		A A B B	4.5 V " " "		IN				OUT						10 to 12 20 to 22 30 to 32 40 to 42	3 " " "	23 " " "	3 " " "	35 " " "	3 " " "	43 " " "	ns " " "
t_{PHL1}	3003 (Fig. 4)			IN		A A B B	4.5 V " " "		IN				OUT						10 to 12 20 to 22 30 to 32 40 to 42	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "
t_{PLH1}	3003 (Fig. 4)			IN		A A B B	4.5 V " " "		IN				OUT						10 to 12 20 to 22 30 to 32 40 to 42	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "
t_{PHL1}	3003 (Fig. 4)			IN		A A B B	4.5 V " " "		IN				OUT						10 to 12 20 to 22 30 to 32 40 to 42	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "
t_{PLH1}	3003 (Fig. 4)			IN		A A B B	4.5 V " " "		IN				OUT						10 to 12 20 to 22 30 to 32 40 to 42	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "
t_{PHL1}	3003 (Fig. 4)			IN		A A B B	4.5 V " " "		IN				OUT						10 to 12 20 to 22 30 to 32 40 to 42	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 test method	Case outline	Terminal conditions																		Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
		Test no.	I7	I3	20	I_{OS}-4	V_{CC}	13	4?	4?	4?	4?	13	4?	4?	3?	GND	16	FCL-2	2?	2?	13	2?								
t_{PLH2}	3003 (Fig. 4)	97		GND	GND		4.5 V		GND								GND	IN	IN			OUT	EN to 17	4	25	4	35	4	25	ns
		98				IN	"										"	"	"				EN to 23	"	"	"	"	"	"	"
		99				IN	"										"	"	"				EN to 33	"	"	"	"	"	"	"
		100					"										"	"	"				EN to 43	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 4)	101	OUT	GND	GND		"						OUT				"	IN	IN	OUT			EN to 17	"	"	"	"	"	"	"
		102					"										"	"	"				EN to 23	"	"	"	"	"	"	"
		103					"		GND	OUT							"	"	"				EN to 33	"	"	"	"	"	"	"
		104					"										"	"	"				EN to 43	"	"	"	"	"	"	"
		105	OUT	4.5 V	4.5 V		"						OUT				"	IN	IN	OUT			EN to 17	"	"	"	"	"	"	"
		106					"										"	"	"				EN to 23	"	"	"	"	"	"	"
		107					"		4.5 V	OUT							"	"	"				EN to 33	"	"	"	"	"	"	"
		108					"										"	"	"				EN to 43	"	"	"	"	"	"	"
t_{PHL}	3004 (Fig. 4)	109		IN	IN		"										"	14.5 V	14.5 V			OUT	EN to 17	"	"	"	"	"	"	"
		110					"										"	"	"				EN to 23	"	"	"	"	"	"	"
		111			IN	14.5 V	"		IN	OUT			OUT				"	"	"				EN to 33	"	"	"	"	"	"	"
		112				14.5 V	"										"	"	"				EN to 43	"	"	"	"	"	"	"
t_{PLH}	3004 (Fig. 4)	113		IN	IN		"										"	14.5 V	14.5 V			OUT	EN to 17	"	"	"	"	"	"	"
		114					"										"	"	"				EN to 23	"	"	"	"	"	"	"
		115			IN	14.5 V	"		IN	OUT			OUT				"	"	"				EN to 33	"	"	"	"	"	"	"
		116				14.5 V	"										"	"	"				EN to 43	"	"	"	"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 32.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case 2 / Case E	Terminal conditions: 1	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits: Subgroup 4 T_C = +25°C Min	Max	Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = -55°C Min	Max	Unit
		Test no.	A0	A1	A2	Q0	Q1	Q2	Q3	GND	Q4	Q5	Q6	Q7	Data IN	EN	CLR	V_{CC}								
C_{in}		67	2/							GND									GND	A0		10					pF
		68		2/						"									"	A1		"					"
		69			2/					"									"	A2		"					"
		70								"					2/				"	Data		"					"
		71								"						2/			"	EN		"					"
		72								"							2/		"	CLR		"					"
																					Subgroup 7 T_C = +25°C		Subgroup 9 T_C = +125°C		Subgroup 8 T_C = -55°C		
Truth table 1/	3014	73	B	B	B	L	L	L	L	GND	L	L	L	L	A	A	B	4.5 V	All Outputs 4/								
		74	B	"	"	H	"	"	"	"	"	"	"	"	"	B	"	"										
		75	A	"	"	L	"	"	"	"	"	"	"	"	"	A	"	"										
		76	A	"	"	"	H	"	"	"	"	"	"	"	"	B	"	"										
		77	B	A	"	"	L	"	"	"	"	"	"	"	"	A	"	"										
		78	B	"	"	"	"	H	"	"	"	"	"	"	"	B	"	"										
		79	A	"	"	"	"	L	"	"	"	"	"	"	"	A	"	"										
		80	A	"	"	"	"	"	H	"	"	"	"	"	"	B	"	"										
		81	B	B	A	"	"	"	L	"	"	"	"	"	"	A	"	"										
		82	B	"	"	"	"	"	"	"	H	"	"	"	"	B	"	"										
		83	A	"	"	"	"	"	"	"	L	"	"	"	"	A	"	"										
		84	A	"	"	"	"	"	"	"	"	"	H	"	"	B	"	"										
		85	B	A	"	"	"	"	"	"	"	"	L	"	"	A	"	"										
		86	B	"	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		87	A	"	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		88	A	"	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		89	A	"	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		90	B	"	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		91	A	B	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		92	B	B	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		93	A	A	B	"	"	"	"	"	"	"	"	"	L	A	"	"										
		94	B	A	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		95	A	B	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		96	B	"	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		97	B	"	"	H	"	"	"	"	"	"	"	"	L	A	"	"										
		98	B	"	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		99	A	"	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		100	A	"	"	"	H	"	"	"	"	"	"	"	H	B	"	"										
		101	A	"	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		102	B	A	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		103	B	"	"	"	"	H	"	"	"	"	"	"	L	A	"	"										
		104	B	"	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		105	A	"	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		106	A	"	"	"	"	"	"	"	H	"	"	"	H	B	"	"										
		107	A	"	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		108	B	B	A	"	"	"	"	"	"	"	"	"	H	B	"	"										
		109	B	"	"	"	"	"	"	"	"	"	H	"	L	A	"	"										
		110	B	"	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		111	A	"	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		112	A	"	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		113	A	"	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		114	B	A	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		115	B	"	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		116	B	"	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
		117	A	"	"	"	"	"	"	"	"	"	"	"	L	A	"	"										
		118	A	"	"	"	"	"	"	"	"	"	"	"	H	B	"	"										
																					Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
t_{PHL1}	3003 (Fig. 4)	119	GND	GND	GND	OUT				GND					IN	GND	4.5 V	4.5 V	Data to Q0		5	37	5	43	5	37	ns
		120	4.5 V	GND	GND		OUT			"					"	5/	"	"	Data to Q1		"	"	"	"	"	"	"
		121	GND	4.5 V	GND			OUT		"					"	"	"	"	Data to Q2		"	"	"	"	"	"	"
		122	4.5 V	4.5 V	GND				OUT	"					"	"	"	"	Data to Q3		"	"	"	"	"	"	"
		123	GND	GND	4.5 V					"	OUT				"	"	"	"	Data to Q4		"	"	"	"	"	"	"
		124	4.5 V	GND	4.5 V					"		OUT			"	"	"	"	Data to Q5		"	"	"	"	"	"	"
		125	GND	4.5 V	4.5 V					"			0.1T		"	"	"	"	Data to Q6		"	"	"	"	"	"	"
		126	4.5 V	4.5 V	4.5 V					"				0.1T	"	"	"	"	Data to Q7		"	"	"	"	"	"	"
t_{PLH1}	3003 (Fig. 4)	127-134	Same tests and terminal conditions and limits as specified above for t_{PHL1}.																								

See footnotes at end of table.

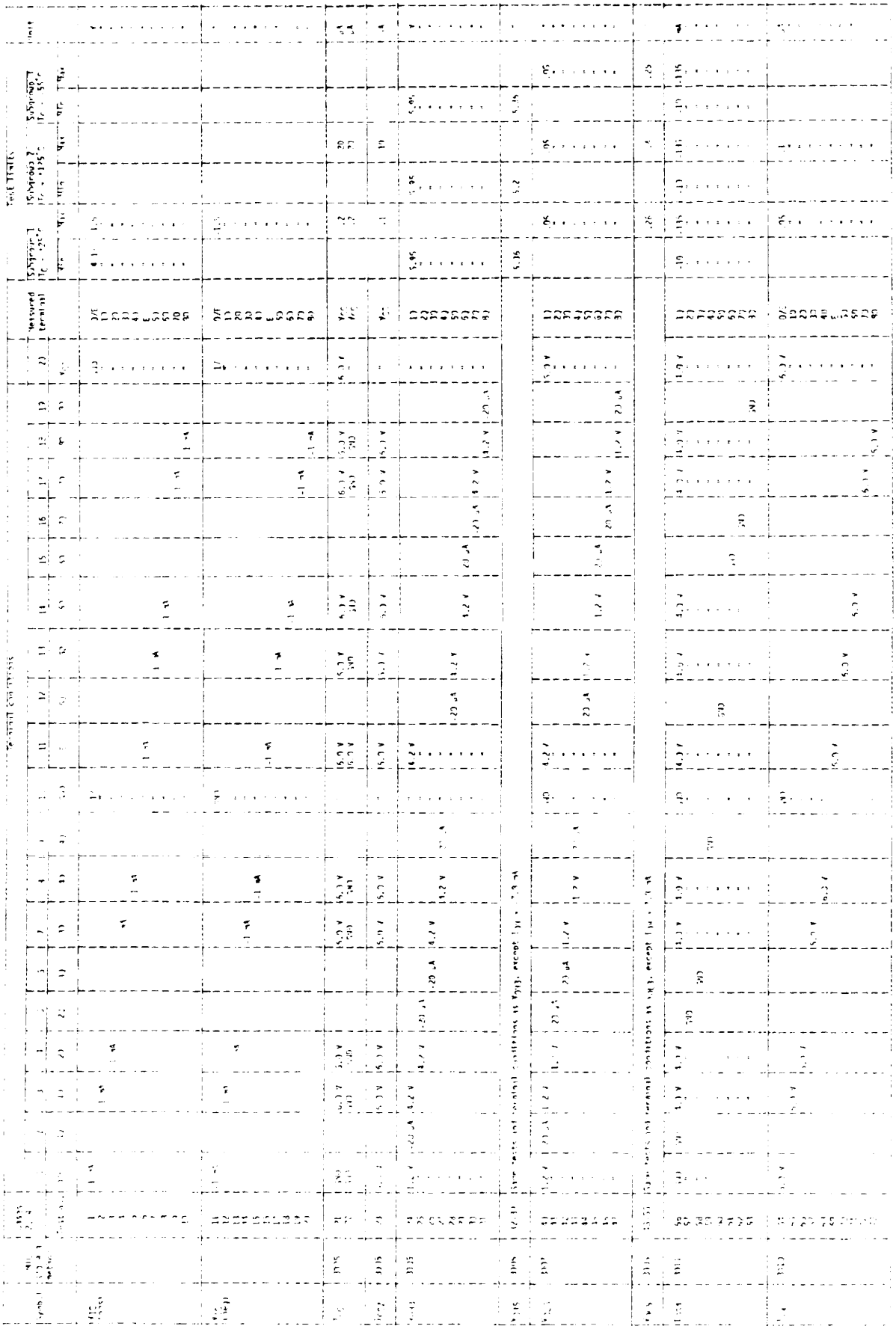
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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case 2 / Case E	Terminal conditions: 2 / 1	3 / 2	4 / 3	5 / 4	7 / 5	9 / 6	9 / 7	10 / 8	12 / 9	13 / 10	14 / 11	15 / 12	17 / 13	18 / 14	19 / 15	20 / 16	Measured terminal	Test limits: Subgroup 9 T_C = +25°C Min	Subgroup 9 T_C = +25°C Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 T_C = +125°C Max	Subgroup 11 T_C = -55°C Min	Subgroup 11 T_C = -55°C Max	Unit
		Test no.	A0	A1	A2	Q3	Q1	Q2	Q3	GND	Q4	Q5	Q6	Q7	Data IN	EN	CLR	V_{CC}								
t_{PHL2}	3003 (Fig. 4)	135	GND	GND	GND	OUT				GND					4.5 V	0 V	4.5 V	4.5 V	EN to Q0	7	35	7	47	7	35	ns
		136	4.5 V	GND	GND		OUT			"					"	"	"	"	EN to Q1	"	"	"	"	"	"	"
		137	GND	4.5 V	GND			OUT		"					"	"	"	"	EN to Q2	"	"	"	"	"	"	"
		138	4.5 V	4.5 V	GND				OUT	"					"	"	"	"	EN to Q3	"	"	"	"	"	"	"
		139	GND	GND	4.5 V					"	OUT				"	"	"	"	EN to Q4	"	"	"	"	"	"	"
		140	4.5 V	GND	4.5 V					"		OUT			"	"	"	"	EN to Q5	"	"	"	"	"	"	"
		141	GND	4.5 V	4.5 V					"			OUT		"	"	"	"	EN to Q6	"	"	"	"	"	"	"
		142	4.5 V	4.5 V	4.5 V					"				OUT	"	"	"	"	EN to Q7	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 4)	143-150	Same tests and terminal conditions and limits as specified above for t_{PHL2}, with the exception that Data IN is at GND.																							
t_{PHL3}	3003	151	IN	GND	GND	OUT				GND					4.5 V	0 V	4.5 V	4.5 V	A0 to Q0	5	39	5	50	5	39	ns
		152	IN	GND	GND		OUT			"					"	"	"	"	A0 to Q1	"	"	"	"	"	"	"
		153	GND	IN	GND			OUT		"					"	"	"	"	A1 to Q2	"	"	"	"	"	"	"
		154	IN	4.5 V	GND				OUT	"					"	"	"	"	A0 to Q3	"	"	"	"	"	"	"
		155	GND	GND	IN					"	OUT				"	"	"	"	A2 to Q4	"	"	"	"	"	"	"
		156	IN	GND	4.5 V					"		OUT			"	"	"	"	A0 to Q5	"	"	"	"	"	"	"
		157	GND	IN	4.5 V					"			OUT		"	"	"	"	A1 to Q6	"	"	"	"	"	"	"
		158	IN	4.5 V	4.5 V					"				OUT	"	"	"	"	A0 to Q7	"	"	"	"	"	"	"
t_{PLH3}	3003 (Fig. 4)	159-166	Same tests and terminal conditions and limits as specified above for t_{PHL3}, with the exception that Data IN is at GND.																							
t_{PHL4}	3003 (Fig. 4)	167	GND	GND	GND	OUT				GND					4.5 V	5/	0 V	4.5 V	Clear to Q0	5	26	5	36	5	25	ns
		168	4.5 V	GND	GND		OUT			"					"	"	"	"	Clear to Q1	"	"	"	"	"	"	"
		169	GND	4.5 V	GND			OUT		"					"	"	"	"	Clear to Q2	"	"	"	"	"	"	"
		170	4.5 V	4.5 V	GND				OUT	"					"	"	"	"	Clear to Q3	"	"	"	"	"	"	"
		171	GND	GND	4.5 V					"	OUT				"	"	"	"	Clear to Q4	"	"	"	"	"	"	"
		172	4.5 V	GND	4.5 V					"		OUT			"	"	"	"	Clear to Q5	"	"	"	"	"	"	"
		173	GND	4.5 V	4.5 V					"			OUT		"	"	"	"	Clear to Q6	"	"	"	"	"	"	"
		174	4.5 V	4.5 V	4.5 V					"				OUT	"	"	"	"	Clear to Q7	"	"	"	"	"	"	"
t_{THL}	3003 (Fig. 4)	175	GND	GND	GND	OUT				GND					0 V	GND	4.5 V	4.5 V	Q0	3	15	3	23	3	15	ns
		176	4.5 V	GND	GND		OUT			"					"	"	"	"	Q1	"	"	"	"	"	"	"
		177	GND	4.5 V	GND			OUT		"					"	"	"	"	Q2	"	"	"	"	"	"	"
		178	4.5 V	4.5 V	GND				OUT	"					"	"	"	"	Q3	"	"	"	"	"	"	"
		179	GND	GND	4.5 V					"	OUT				"	"	"	"	Q4	"	"	"	"	"	"	"
		180	4.5 V	GND	4.5 V					"		OUT			"	"	"	"	Q5	"	"	"	"	"	"	"
		181	GND	4.5 V	4.5 V					"			OUT		"	"	"	"	Q6	"	"	"	"	"	"	"
		182	4.5 V	4.5 V	4.5 V					"				OUT	"	"	"	"	Q7	"	"	"	"	"	"	"
t_{TLH}	3003 (Fig. 4)	183-190	Same tests and terminal conditions and limits as specified above for as t_{THL}.																							

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TABLE III. Group A inspection for device type 03 - Continued.

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 4	Terminal conditions 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Subgroup 1 (T_C = +25°C) Min	Max	Subgroup 2 (T_C = +125°C) Min	Max	Subgroup 3 (T_C = -55°C) Min	Max	Unit
		Test no.																												
I_{CCH1}	3003 (Fig. 4)	125	GND									GND																		
		126																												
		127																												
		128																												
		129																												
		130																												
		131																												
		132																												
I_{CCH2}	3003 (Fig. 4)	(133-144)	Same tests and terminal conditions and limits as specified above for I_{CCH1}.																											
I_{CCH2}	3003 (Fig. 4)	141	GND	OUT	14.5 V	14.5 V	NIC	OUT	14.5 V	14.5 V	OUT	GND	14.5 V	OUT	14.5 V	14.5 V	OUT	14.5 V	OUT	14.5 V	14.5 V	OUT	14.5 V	E to 10	6	33	6	41	6	31	mA
		142																						E to 20							
		143																						E to 30							
		144																						E to 40							
		145																						E to 50							
		146																						E to 60							
		147																						E to 70							
		148																						E to 80							
I_{CCH2}	3003 (Fig. 4)	149	GND	OUT	GND	GND	OUT	OUT	GND	GND	OUT	GND	14.5 V	OUT	GND	GND	OUT	GND	OUT	GND	GND	OUT	14.5 V	E to 10	6	33	6	41	6	31	mA
		150																						E to 20							
		151																						E to 30							
		152																						E to 40							
		153																						E to 50							
		154																						E to 60							
		155																						E to 70							
		156																						E to 80							
I_{CCH2}	3003 (Fig. 4)	157	IN	OUT	14.5 V	14.5 V	OUT	OUT	14.5 V	14.5 V	OUT	GND	14.5 V	OUT	14.5 V	14.5 V	OUT	14.5 V	OUT	14.5 V	14.5 V	OUT	14.5 V	E to 10	5	24	5	35	5	24	mA
		158																						E to 20							
		159																						E to 30							
		160																						E to 40							
		161																						E to 50							
		162																						E to 60							
		163																						E to 70							
		164																						E to 80							
I_{CCL7}	3003 (Fig. 4)	165	IN	OUT	GND	GND	OUT	OUT	GND	GND	OUT	GND	14.5 V	OUT	GND	GND	OUT	GND	OUT	GND	GND	OUT	14.5 V	E to 10	6	33	6	41	6	31	mA
		166																						E to 20							
		167																						E to 30							
		168																						E to 40							
		169																						E to 50							
		170																						E to 60							
		171																						E to 70							
		172																						E to 80							
I_{CCL2}	3003 (Fig. 4)	173	IN	OUT	14.5 V	14.5 V	OUT	OUT	14.5 V	14.5 V	OUT	GND	14.5 V	OUT	14.5 V	14.5 V	OUT	14.5 V	OUT	14.5 V	14.5 V	OUT	14.5 V	E to 10	5	24	5	35	5	24	mA
		174																						E to 20							
		175																						E to 30							
		176																						E to 40							
		177																						E to 50							
		178																						E to 60							
		179																						E to 70							
		180																						E to 80							
I_{CCL2}	3003 (Fig. 4)	(181-194)	Same tests and terminal conditions and limits as specified above for I_{CCL1}.																											

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 3, 4	Terminal conditions: 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits: Subgroup 9 TC = +25°C		Subgroup 10 TC = +125°C		Subgroup 11 TC = -55°C		Unit
																								Min	Max	Min	Max	Min	Max	
ITH	3004	189																					10	2	12	2	16	2	12	ns
	Fig. 4	190																					20							
		191																					30							
		192																					40							
		193																					50							
		194																					60							
		195																					70							
		196																					80							
ITHH	3004	197-204	Same tests and terminal conditions and limits as specified above for ITH.																											
	Fig. 4																													

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TABLE III. Group A inspection for device type 01.

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 4 Test no.	Terminal conditions 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = +125°C Min	Max	Subgroup 9 T_C = -55°C Min	Max	Limit
Truth table 3/	3014	119	R	L	A	B	H	L	A	B	H	GND	A	L	A	B	H	L	A	B	H	4.5 V	All outputs						
		120		H	B	A	L	H	B	A	L		B	H	B	A	L	H	B	A	L		4/							
		121		H	A	B	H	L	A	B	L		A	H	A	B	H	L	A	B	L									
		122		H	B	B	L	H	B	B	L		B	L	B	A	H	L	B	A	H									
		123		H	A	A	H	L	A	A	L		A	H	A	B	L	H	A	B	H									
		124		H	B	A	L	H	B	A	L		B	L	B	A	L	H	B	A	H									
		125		L	A	B	H	L	A	B	H		A	L	A	B	L	H	A	B	H									
		126		L	B	B	H	L	B	B	H		B	L	B	A	L	H	B	A	H									
		127		L	A	A	H	L	A	A	H		A	L	A	B	L	H	A	B	H									
																								Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
t_{PHL1}	3003 (Fig. 4)	128	0.0 V	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	14.5 V	OUT	IN	IN	OUT	OUT	IN	IN	OUT	14.5 V	00 to 07	6	29	6	35	6	29	ns
		129																					01 to 07							
		130																					02 to 07							
		131																					03 to 07							
		132																					04 to 07							
		133																					05 to 07							
		134																					06 to 07							
		135																					07							
t_{PHL1}	3003 (Fig. 4)	136-143	Same tests, terminal conditions, and limits as specified above for t_{PLH1}.																											
t_{PLH2}	3003 (Fig. 4)	144	0.0 V	OUT	10.0 V	10.0 V	OUT	OUT	10.0 V	10.0 V	OUT	GND	IN	OUT	10.0 V	0.0 V	OUT	OUT	10.0 V	10.0 V	OUT	4.5 V	00 to 07	6	31	6	41	6	31	ns
		145																					01 to 07							
		146																					02 to 07							
		147																					03 to 07							
		148																					04 to 07							
		149																					05 to 07							
		150																					06 to 07							
		151																					07							
t_{PHL2}	3003 (Fig. 4)	152	0.0 V	OUT	14.5 V	14.5 V	OUT	OUT	14.5 V	14.5 V	OUT	GND	IN	OUT	14.5 V	4.5 V	OUT	OUT	14.5 V	14.5 V	OUT	4.5 V	00 to 07							
		153																					01 to 07							
		154																					02 to 07							
		155																					03 to 07							
		156																					04 to 07							
		157																					05 to 07							
		158																					06 to 07							
		159																					07							
t_{PHZ}	3003 (Fig. 4)	160	IN	OUT	0.0 V	0.0 V	OUT	OUT	10.0 V	10.0 V	OUT	GND	14.5 V	OUT	0.0 V	0.0 V	OUT	OUT	0.0 V	10.0 V	OUT	4.5 V	00 to 07	5	26	5	33	5	26	ns
		161																					01 to 07							
		162																					02 to 07							
		163																					03 to 07							
		164																					04 to 07							
		165																					05 to 07							
		166																					06 to 07							
		167																					07							
t_{PLZ}	3003 (Fig. 4)	168	IN	OUT	14.5 V	14.5 V	OUT	OUT	14.5 V	14.5 V	OUT	GND	14.5 V	OUT	14.5 V	4.5 V	OUT	OUT	14.5 V	14.5 V	OUT	4.5 V	00 to 07							
		169																					01 to 07							
		170																					02 to 07							
		171																					03 to 07							
		172																					04 to 07							
		173																					05 to 07							
		174																					06 to 07							
		175																					07							

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TABLE III. Group A inspection for device type 31 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 8	Test no.	Terminal conditions: 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits: Subgroup 8 T_C = +25°C Min	Subgroup 8 T_C = +25°C Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 T_C = +125°C Max	Subgroup 11 T_C = -55°C Min	Subgroup 11 T_C = -55°C Max	Unit
t_{PLH}	3003 (Fig. 41)		176	IN	OUT	0.0 V	0.0 V	OUT	OUT	0.0 V	0.0 V	GND	14.5 V		OUT	0.0 V		OUT					14.5 V	BE, B0, B1, B2, B3, B4, B5, B6, B7	5	20	5	35	5	25	ns
			177																												
			178																												
			179																												
			180																												
			181																												
			182																												
			183																												
t_{PHL}	3003 (Fig. 41)		184	IN	OUT	14.5 V	14.5 V	OUT	OUT	14.5 V	14.5 V	GND	14.5 V		OUT	14.5 V		OUT					0.0 V	BE, B0, B1, B2, B3, B4, B5, B6, B7	2	12	2	16	2	12	ns
			185																												
			186																												
			187																												
			188																												
			189																												
			190																												
			191																												
t_{PHL}	3004 (Fig. 41)		192	10.0 V	OUT	IN	IN	OUT	OUT	IN	IN	GND	14.5 V		OUT	IN		OUT					0.0 V	BE, B0, B1, B2, B3, B4, B5, B6, B7	2	12	2	16	2	12	ns
			193																												
			194																												
			195																												
			196																												
			197																												
			198																												
			199																												
t_{PLH}	3004 (Fig. 41)		200-207	Same tests, terminal conditions, and limits as specified above for t_{PHL}																											

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TABLE III. Group A inspection for device type 03.

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TABLE III. Group A inspection for device type 35 - Continued.

Symbol	MIL-STD-883 test method	Cases C, D	Test no.	Terminal conditions	Measured terminal	Limits Subgroup 1 T_C = +25°C	Limits Subgroup 2 T_C = +125°C	Limits Subgroup 3 T_C = -55°C	Unit
						Min / Max	Min / Max	Min / Max	
I_{CCH}	3011	56	1	GND	V_{CC}		10		mA
		57	2	GND					
		58	3	GND					
		59	4	GND					
		60	5	GND					
		61	6	GND					
		62	7	GND					
		63	8	GND					
I_{CCL}	3005	64	1	4.5 V	V_{CC}		135		mA
		65	2	GND					
		66	3	GND					
		67	4	GND					
		68	5	GND					
		69	6	GND					
		70	7	GND					
		71	8	GND					
I_{CCL}	3005	72	1	GND	V_{CC}		135		mA
		73	2	GND					
		74	3	GND					
		75	4	GND					
		76	5	GND					
		77	6	GND					
		78	7	GND					
		79	8	GND					
I_{IH}	3010	80	1	4.5 V	A0		1		µA
		81	2	GND	A1				
		82	3	GND	A2				
		83	4	GND	D0				
		84	5	GND	D1				
		85	6	GND	D2				
		86	7	GND	D3				
		87	8	GND	D4				
		88	9	GND	D5				
		89	10	GND	D6				
		90	11	GND	D7				
		91	12	GND	LE				
I_{IH}	3010	92	1	GND	A0		1		µA
		93	2	GND	A1				
		94	3	GND	A2				
		95	4	GND	D0				
		96	5	GND	D1				
		97	6	GND	D2				
		98	7	GND	D3				
		99	8	GND	D4				
		100	9	GND	D5				
		101	10	GND	D6				
		102	11	GND	D7				
		103	12	GND	LE				
I_{IH}	3010	104	1	GND	A0		1		µA
		105	2	GND	A1				
		106	3	GND	A2				
		107	4	GND	D0				
		108	5	GND	D1				
		109	6	GND	D2				
		110	7	GND	D3				
		111	8	GND	D4				
		112	9	GND	D5				
		113	10	GND	D6				
		114	11	GND	D7				
		115	12	GND	LE				

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 Method	Cases 2, 8	Terminal conditions																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 9 T_C = -55°C		
		Test no	$\overline{OE}$	D3	D1	D2	D3	D4	D5	D6	D7	GND	$\overline{EE}$	Q7	Q6	Q5	Q4	Q3	Q2	Q1	Q0	V_{CC}		Min	Max	Min	Max	Min	Max	
Truth table 3/	3014	118 119 120 121 122 123 124 125 126	B A B A A A B A A	A B B A A A B A A	B A A B B B A B B	A B A B A A B A A	B A B A B B A B B	A B A B A A B A A	B A B A B B A B B	A B A B A A B A B	B A A B B B A B B	GND	A A B B A A B A B	H L L L H L H H H	L H H H L H L L L	H L L L H L H H H	L H H H L H L L L	H L L L H L H H H	L H H H L H L L L	H L L L H L H H H	L H H H L H L L L	14.5 V	8/							
																								Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
t_{PLH1}	3003 (Fig. 4)	127 128 129 130 131 132 133 134	0.0 V	IN	IN	IN	IN	IN	IN	IN	IN	GND	4.5 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	4.5 V	$\overline{OE}$ to Q0 D0 to Q0 D1 to Q1 D2 to Q2 D3 to Q3 D4 to Q4 D5 to Q5 D6 to Q6 D7 to Q7	4	31	4	41	4	31	ns
t_{PHL1}	3003 (Fig. 4)	135 142	Same tests, terminal conditions and limits as specified above for t_{PLH1}																											
t_{PLH2}	3003 (Fig. 4)	143 144 145 146 147 148 149 150	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	GND	IN	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	4.5 V	$\overline{OE}$ to Q0 D0 to Q0 D1 to Q1 D2 to Q2 D3 to Q3 D4 to Q4 D5 to Q5 D6 to Q6 D7 to Q7	4	31	4	41	4	31	ns
t_{PHL2}	3003 (Fig. 4)	151 152 153 154 155 156 157 158		4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	GND	IN	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	4.5 V	$\overline{OE}$ to Q0 D0 to Q0 D1 to Q1 D2 to Q2 D3 to Q3 D4 to Q4 D5 to Q5 D6 to Q6 D7 to Q7							
t_{PHZ}	3003 (Fig. 4)	159 160 161 162 163 164 165 166	IN	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	GND	4.5 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	4.5 V	$\overline{OE}$ to Q0 D0 to Q0 D1 to Q1 D2 to Q2 D3 to Q3 D4 to Q4 D5 to Q5 D6 to Q6 D7 to Q7	3	26	3	35	3	26	ns
t_{PLZ}	3003 (Fig. 4)	167 168 169 170 171 172 173 174		4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	GND	4.5 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	4.5 V	$\overline{OE}$ to Q0 D0 to Q0 D1 to Q1 D2 to Q2 D3 to Q3 D4 to Q4 D5 to Q5 D6 to Q6 D7 to Q7							

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TABLE II-I. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 3	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test Limits Subgroup 9 TC = +25°C Min	Test Limits Subgroup 9 TC = +25°C Max	Test Limits Subgroup 10, 11 TC = +125°C, -55°C Min	Test Limits Subgroup 10, 11 TC = +125°C, -55°C Max	Unit
		Test No.	$\overline{CE}$	D1	D2	D3	D4	D5	D6	D7	GND	$\overline{CE}$	$\overline{Y7}$	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}							
t_{PLH}	3003 (Fig. 4)	175		0.0 V								4.5 V				OUT					4.5 V							ns
		176			0.0 V																							
		177				0.0 V																						
		178					0.0 V																					
		179						0.0 V																				
		180							0.0 V																			
		181								0.0 V																		
		182									0.0 V																	
t_{PHL}	3003 (Fig. 4)	183		4.5 V								4.5 V				OUT					4.5 V							ns
		184			4.5 V																							
		185				4.5 V																						
		186					4.5 V																					
		187						4.5 V																				
		188							4.5 V																			
		189								4.5 V																		
		190									4.5 V																	
t_{PLH}	3004 (Fig. 5)	191		IN								OUT																ns
		192			IN																							
		193				IN																						
		194					IN																					
		195						IN																				
		196							IN																			
		197								IN																		
		198									IN																	
t_{PLH}	3004 (Fig. 5)	199 thru 206	Same tests, terminal conditions and limits as specified above for t_{PLH}.																									

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TABLE III. Group A inspection for device type 06.

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TABLE III. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Test limits Subgroup 1 T_C = +25°C Max	Test limits Subgroup 2 T_C = +125°C Min	Test limits Subgroup 2 T_C = +125°C Max	Test limits Subgroup 3 T_C = -55°C Min	Test limits Subgroup 3 T_C = -55°C Max	Unit
		Test no.	O/I	I/O	I/O	I/O	I/O	I/O	I/O	I/O	I/O	I/O	EMC	EMC	EMC	EMC	EMC	EMC	EMC	EMC	EMC	V_{CC}								
				GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND																
I_{IL}	3009	78																					10		-.05					mA
		75																					70							
		76																					30							
		77																					40							
		78																					50							
		79																					60							
		93																					70							
		81																					80							
		82																					D/E							
		83																					EMC							
I_{IH1}		84																					80		2.0					
		85																					70							
		86																					60							
		87																					50							
		88																					40							
		89																					30							
		90																					20							
		91																					10							
I_{IH2}		92																					80		2.0					
		93																					70							
		94																					60							
		95																					50							
		96																					40							
		97																					30							
		98																					20							
		99																					10							
I_{OS}	3011	100																					O/I		pF					
		101																					10							
		102																					70							
		103																					30							
		104																					40							
		105																					50							
		106																					60							
		107																					70							
		108																					80							
		109																					EMC							
I_{CC}	3012	110	16.0 V										16.0 V										80		23					
		111																					70							
		112																					60							
		113																					50							
		114																					40							
		115																					30							
		116																					20							
		117																					10							

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TABLE III. Group A inspection for device type 05 - Continued.

| Symbol | MIL-STD-883 method | Cases | Test no. | | |

TABLE III. Group A inspection for device type 06 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 3	Terminal conditions																			Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Subgroup 4 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 6 T_C = -55°C	
		Test no.	D/E																					Min	Max	Min	Max	Min	Max	
t_{PZL}	3003 (Fig. 4)	184	IN	GND	GND	GND	GND	GND	GND	GND	GND	GND	4.5 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	V_{CC}	1C to 1O 2C to 2O 3C to 3O 4C to 4O 5C to 5O 6C to 6O 7C to 7O 8C to 8O	4	26	4	35	4	26	ns
		185	-																											
		186	-																											
		187	-																											
		188	-																											
		189	-																											
		190	-																											
		191	-																											
t_{PHL}	3004 (Fig. 4)	192	GND	IN										OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	V_{CC}	1O 2O 3O 4O 5O 6O 7O 8C	2	12	2	16	2	12	ns
		193	-																											
		194	-		IN	IN	IN	IN	IN	IN	IN	IN																		
		195	-																											
		196	-																											
		197	-																											
		198	-																											
		199	-																											
t_{PLH}	3004 (Fig. 4)	200-207	Same tests, terminal conditions and limits as specified above for t_{PHL}.																											

See footnotes at end of table.

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TABLE III. Group A inspection for device type 51

Symbol	MIL-STD-883 method	Cases 2, 8	Terminal conditions: 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits: Subgroup 1 T_C = +25°C min	Subgroup 1 T_C = +25°C max	Subgroup 2 T_C = +125°C min	Subgroup 2 T_C = +125°C max	Subgroup 3 T_C = -55°C min	Subgroup 3 T_C = -55°C max	Unit
		Test no.	I/L	I/L	I/O	I/O	I/O	I/O	I/O	I/O	I/O	GND	E	SO	SO	SO	SO	I/O	I/O	RO	RO	V_{CC}								
V_{IC} (pos)		1	1 mA		1 mA							I/										GND	O/E	.4 1/	1.5					V
		2				1 mA						"										"	I/O	"	"					"
		3										"										"	I/O	"	"					"
		4							1 mA			"										"	I/O	"	"					"
		5								1 mA		"										"	I/O	"	"					"
		6										"	1 mA									"	E	"	"					"
		7										"										"	SO	"	"					"
		8										"			1 mA	1 mA						"	SO	"	"					"
		9										"							1 mA			"	I/O	"	"					"
		10										"								1 mA			"	RO	"	"					"
V_{IC} (neg)		11	-1 mA		-1 mA							GND										1/	O/E		-1.5					V
		12				-1 mA						"										"	I/O		"					"
		13							-1 mA			"										"	I/O		"					"
		14								-1 mA		"										"	I/O		"					"
		15										"	-1 mA									"	E		"					"
		16										"			-1 mA	-1 mA						"	SO		"					"
		17										"							-1 mA			"	I/O		"					"
		18										"								-1 mA			"	RO		"					"
		19										"									-1 mA		"	I/O		"					"
		20										"			-1 mA							"	RO		"					"
I_{CC}	3005	21	GND		5.5 V	5.5 V			5.5 V	5.5 V		"	5.5 V		5.5 V	5.5 V			5.5 V	5.5 V		5.5 V	V_{CC}		.2		20			mA
		22	GND		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	V_{CC}		.2		23			"
I_{CCZ}	3005	23	5.5 V		5.5 V	5.5 V			5.5 V	5.5 V		"	"		5.5 V	5.5 V			5.5 V	5.5 V		"	V_{CC}		.1		15			"
I_{CCS}	3005	24	12.6 V									"										"	V_{CC}		.735		1.5			mA
		25	10.8 V									"	12.4 V									"	V_{CC}		.295		"			"
		26	10.8 V		10.8 V	10.8 V			10.8 V	10.8 V		"	5.5 V		10.8 V	0.8 V			10.8 V	12.4 V		"	V_{CC}		.195		"			"
I_{OZH}		27	7/	5.5 V	7/							"	5.5 V									"	I/O		.2		2.0			mA
		28	"			7/	5.5 V					"	"									"	I/O		"		"			"
		29	"					5.5 V	7/			"	"									"	I/O		"		"			"
		30	"							7/	5.5 V	"	"									"	I/O		"		"			"
		31	"									"	"	5.5 V	7/	7/						"	I/O		"		"			"
		32	"									"	"									"	I/O		"		"			"
		33	"									"	"									"	I/O		"		"			"
		34	"									"	"									"	I/O		"		"			"
I_{OZL}		35	"	GND	7/							"										"	I/O		-.2		-2.0			mA
		36	"			7/	GND					"	"									"	I/O		"		"			"
		37	"					GND	7/			"	"									"	I/O		"		"			"
		38	"							7/	GND	"	"									"	I/O		"		"			"
		39	"									"	"	GND	7/	7/						"	I/O		"		"			"
		40	"									"	"									"	I/O		"		"			"
		41	"									"	"									"	I/O		"		"			"
		42	"									"	"									"	I/O		"		"			"
V_{OH6}	3006	43	10.9 V	-20 µA	12.0 V							"	12.0 V									4.5 V	I/O	9.4		9.4		9.4		V
		44	"			12.0 V	-20 µA					"	"									"	I/O	"		"		"		"
		45	"					-20 µA	12.0 V			"	"									"	I/O	"		"		"		"
		46	"							12.0 V	-20 µA	"	"									"	I/O	"		"		"		"
		47	"									"	"	-20 µA	12.0 V	2.0 V						"	I/O	"		"		"		"
		48	"									"	"									"	I/O	"		"		"		"
		49	"									"	"									"	I/O	"		"		"		"
V_{OH7}	3006	50-58	Same tests and terminal conditions as V_{OH6}, except I_{OH} = -5.0 mA.																			5.5 V		3.98		3.7		3.98		V
V_{OL6}	3007	59	10.8 V	20 µA	10.8 V							"	12.0 V									4.5 V	I/O		.1		.1		.1	V
		60	"			10.8 V	20 µA					"	"									"	I/O		"		"		"	"
		61	"					20 µA	10.8 V			"	"									"	I/O		"		"		"	"
		62	"							10.8 V	20 µA	"	"									"	I/O		"		"		"	"
		63	"									"	"	20 µA	10.8 V	0.8 V						"	I/O		"		"		"	"
		64	"									"	"									"	I/O		"		"		"	"
		65	"									"	"									"	I/O		"		"		"	"
		66	"									"	"									"	I/O		"		"		"	"
V_{OL7}	3007	67-74	Same tests and terminal conditions as V_{OL6}, except I_{OL} = 5.0 mA.																			5.5 V			.25		.4		.25	V

See footnotes at end of table.

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TABLE III. Group A inspection for device type 53 - Continued.

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TABLE III. Group A inspection for device type 03 - Continued.

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	1/	1Q	1D	2D	2C	3Q	3D	4D	4Q	5Q	5	6Q	6	7Q	7D	8D	8Q	9Q	9D	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PZH}	3003 (Fig. 4)	176	1Q	OUT	4.5 V		OUT					GND	4.5 V									4.5 V	1Q to 1J	5	31	5	41	5	34	ns
		177	"			4.5 V						"	"									"	2Q to 2J	"	"	"	"	"	"	"
		178	"					OUT	4.5 V			"	"									"	3Q to 3J	"	"	"	"	"	"	"
		179	"							4.5 V	OUT	"	"									"	4Q to 4J	"	"	"	"	"	"	"
		180	"									"	"	OUT	4.5 V							"	5Q to 5J	"	"	"	"	"	"	"
		181	"									"	"			4.5 V	OUT					"	6Q to 6J	"	"	"	"	"	"	"
		182	"									"	"					OUT	4.5 V			"	7Q to 7J	"	"	"	"	"	"	"
		183	"									"	"							4.5 V	OUT	"	8Q to 8J	"	"	"	"	"	"	"
t_{PZL}	3003 (Fig. 4)	184-191	Same tests, terminal conditions and limits as specified above for t_{PZH}.																											
t_{THL}	3004 (Fig. 4)	192	GND	OUT	IN	IN	OUT					GND	4.5 V									4.5 V	1Q	2	12	2	16	2	12	ns
		193	"					OUT	IN			"	"									"	2Q	"	"	"	"	"	"	"
		194	"							IN	OUT	"	"	OUT	IN							"	3Q	"	"	"	"	"	"	"
		195	"									"	"			IN	OUT					"	4Q	"	"	"	"	"	"	"
		196	"									"	"					OUT	IN			"	5Q	"	"	"	"	"	"	"
		197	"									"	"							IN	OUT	"	6Q	"	"	"	"	"	"	"
		198	"									"	"									"	7Q	"	"	"	"	"	"	"
		199	"									"	"									"	8Q	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 4)	200-208	Same tests, terminal conditions and limits as specified above for t_{THL}.																											

1/ Input pins not designated shall be "high" level logic or "low" level logic, or may be left open provided they do not influence the outcome of the measurement. Output pins not designated shall be tied to the loads or left open provided they do not influence the outcome of the measurement.
Exceptions are as follows:

a. V_{IC} (pos) tests, the GND terminal shall be open and the minimum limit of 0.4 V applies only where test equipment limitations do not allow for the GND pin to be open during test.
b. V_{IC} (neg) tests, the V_{CC} terminal shall be open.
c. I_{CC} tests, the output terminals shall be open.

2/ See 4.4.1c.

3/ Tests shall be performed in sequence, attributes data only.

4/ H ≥ 2.5 V; L ≤ 2.5 V; A = 3.7 V, B = 0.4 V, except for device 53; A = 2.4 V; B = 0.4 V.

5/ EN must be held at logic "1" while changing the address lines (A_0, A_1, A_2) to select the next output bit.

6/ Apply 4.5 V —— = 4.5 V
0 V |_____| momentary
pulse prior to each test.

7/ Three-state output conditions are required for I_{OZL}, set output to high state. For I_{OZH}, set output to low state. Set input pins to V_{IL} = V_{IL} (max) and V_{IH} = V_{IH} (min), as required.

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4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within 10°C of their power stable condition at room temperature; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at 25°C.

Parameter 1/	Device types	
	All	
I_{CC}		±30 nA

1/ The above parameter shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

4.5.3 Quiescent supply current (I_{CC} test). When performing quiescent supply current measurements (I_{CC}), the meter shall be placed so that all currents flow through the meter.

4.5.4 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table V in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end point electrical parameters as defined in table IV at 25°C, after exposure. The start and completion of the end point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table V and figure 5. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

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TABLE V. Threshold-voltage test circuit conditions.

Device type	GND	5 V	V_{IN} measured at -10 µA supply	GND	-5.0 V	V_{TP} measured at 10 µA supply
01	12	5	2,3,4,6,7,13	12	2,3,4,6,7,13	5
02	8	16	1-3,13,14,15	8	1-3,13,14,15	16
03 53	10	20	1,3,4,7,8,11,13, 14,17,18	20	1,3,4,7,8,11,13, 14,17,18	20
04	10	20	1,3,4,7,8,11,13, 14,17,18	10	1,3,4,7,8,11,13, 14,17,18	20
05	10	20	1-9,11	10	1-9,11	20
06	10	20	1-9,11	10	1-9,11	20

TABLE VI. Radiation hardening end-point electrical parameters at +25°C.

Parameter	All device types	V_{CC} Device types 01-06	V_{CC} Device type 53
V_{TN}	-0.3 V min	5.0 V	5.0 V
V_{TP}	2.8 V max	5.0 V	5.0 V
ΔV_T	1.0 V max	5.0 V	5.0 V
I_{CC}	100 x max limit	6.0 V	5.5 V
t_{PLH}	1.35 x max limit	4.5 V	4.5 V
t_{PHL}	1.35 x max limit	4.5 V	4.5 V

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TABLE VII. Threshold-voltage test circuit conditions.

pe	Pin connections		
	V_{CC} = 4.5 V dc (through a 30- to 60-kilohm resistor)	GND	V_{CC} = 4.5 V dc
	2, 3, 4, 6, 7, 13	12	5
	1-3, 13-15	8	16
3	1, 3, 4, 7, 8, 11, 13, 14, 17, 18	10	20
	1, 3, 4, 7, 8, 11, 13, 14, 17, 18	10	20
	1-9, 11	10	20
	1-9, 11	10	20

esignated are open or connected to 4.5 V dc through a 30- to 60-kilohm

a reporting. When specified in the purchase order or contract, a copy of ing data, as applicable, shall be supplied.

tributes data for all screening tests (see 4.4) and variables data for all tatic burn-in, dynamic burn-in, and steady-state life tests (see 3.5).

copy of each radiograph.

ne quality conformance inspection data (see 4.4).

ar ter distribution data on parameters evaluated during burn-in (see 3.5).

inal electrical parameters data (see 4.2c).

AGING

kaging requirements. The requirements for packaging shall be in accordance -38510.

S

ended use. Microcircuits conforming to this specification are intended for quipment design application and logistic support of existing equipment.

ring data. The acquisition document should specify the following:

mplete part number (see 1.2).

equirements for delivery of one copy of the quality conformance inspection ta pertinent to the device inspection lot to be supplied with each hipment by the device manufacturer, if applicable.

equirements for certificate of compliance, if applicable.

equirements for notification of change of product or process to the ontracting activity in addition to notification to the qualifying activity, if applicable.

equirements for failure analysis (including required test condition of ethod 5003 of MIL-STD-883), corrective action and reporting of results, if pplicable.

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f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by, or direct shipment to the Government.

h. Requirements for "JAN" marking.

i. Requirements for total dose radiation testing (see 3.6.1 and 4.5.4), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-883, and as follows:

C_i	Input terminal-to-GND capacitance.
GND	Ground zero voltage potential.
I_{CC}	Quiescent supply current.
T_A	Free air temperature.
V_{CC}	Positive supply voltage.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54HC75
02	54HC259
03	54HC373
04	54HC533
05	54HC563
06	54HC573
53	54HCT373

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static build up. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

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ans:
- ER
- EC
orce - 17
- NA

activities:
- AR, MI
orce - 11, 19, 85, 99
- ES

tivities:
- SM
- AS, CG, OC, MC, SH

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0953)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/654	Microcircuits, Digital, High Speed, CMOS Latches, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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